Experimental Techniques to Measure Transport Properties

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Abstract

Transport measurements are one of the pillars of experimental condensed matter research, using which several landmark discoveries (eg, superconductivity) have been made. In this teaching lab, we introduce the student to current-voltage transport measurements under controlled thermodynamic conditions and the techniques involved. We highlight various sources of error, and how specific techniques are used to overcome them. We emphasize and illustrate the strong interplay between theoretical modelling and experimental data throughout the course of the experiments.

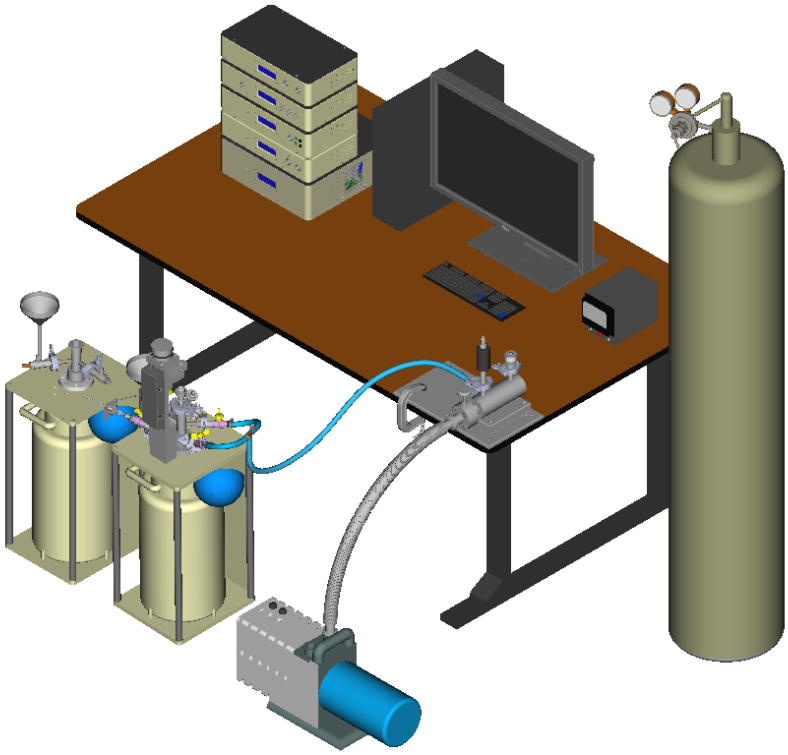


FIG. 1. `XPLORE` transport measurement system

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I. INTRODUCTION

A. Current-Voltage characteristics

Current-Voltage (I-V) relationships are one of the fundamental defining properties of a material. A *conductor* allows for the passage of a finite charge current I given an applied potential difference V. An *insulator* resists a charge flow $I = 0$ for a large range of V. A *superconductor* on the other hand permits a finite current with no associated potential drop $V = 0$. We list these three broad classes of materials based on their I-V characteristics in Table I

Material	I	V
Conductor	finite	finite
Insulator	0	finite
Superconductor	finite	0

TABLE I. Classes of materials based on their IV characteristics

1. Theory → Experiment

One approach to studying the IV characteristics of a material is to first model the dynamics of charge transport in the material, and then *compute* the IV curves. Experiments are then used to check if the theory predicts the right behavior (eg, $I \propto V$ or $I \propto V^2$), as well as to fix the unknown parameters of the theory (eg, the proportionality constant R in $I = V/R$).

If the measurements do not match the theoretical calculations, attempts are made to either refine the theory by relaxing certain assumptions made in the derivation of the original theory, or develop a more sophisticated model that is able to explain the measurements. This approach is driven by theory, and uses experiments to corroborate the theoretical models. The emphasis in this approach is on the derivation of the theory, and the mathematical and computational techniques to calculate the IV curves from the theory. This approach is what is typically followed in a theoretical course in condensed matter physics.

2. Experiment → Theory

The approach followed in this course is on first setting up experiments to perform IV measurements of various materials. The emphasis is on the use of various experimental techniques to allow

for accurate measurements and to develop the technical skills needed to use an assorted range of electronic, cyrogenic and vacuum systems which are needed to make the measurements. At every step of the measurement process, the errors are quantified and understood. Once there is control over the experimental process, the physical system can be probed in various ways (eg, how does R vary with temperature) in order to test known theoretical models and develop an understanding of the charge transport mechanisms in the material.

B. Techniques in Experimental Physics

We attempt to illustrate techniques in experimental physics by considering an example in theoretical physics. Consider the Schrödinger equation that describes the time evolution of the wave function $\psi(\mathbf{r}, t)$ of a non-relativistic particle with mass m moving in a potential field $V(\mathbf{r}, t)$,

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi \tag{1}$$

A typical first course in quantum mechanics involves solving the above equation for set of restricted time independent potentials $V(\mathbf{r})$, such as an infinite square well, a finite square well, and the $1/r$ Coulomb potential leading to the stationary states of the Hydrogen atom. This involves learning the technique of solving for the wavefunction $\psi(\mathbf{r})$ by expanding in a series of special functions subject to the symmetries of the problem. When the solution is an infinite series, it is of paramount importance to check whether the series is convergent and to note the error that arises due to finite truncation.

To go further, various perturbative techniques from mathematical physics are used to include the effects of electric and magnetic fields on the atomic spectrum of Hydrogen. In applying these techniques, the errors involved in the approximations are noted and presented on equal footing as the solution.

When the perturbative techniques break down, or when there are no special symmetries that can be exploited to solve the problem analytically, one must resort to numerical techniques that are typically taught in a course on computational physics. A representation of (1) is put on a computer and evolved using various numerical schemes. The numerical solutions obtained need to be checked against various analytic and semi-analytic solutions of (1). Understanding and reporting the errors in the solutions produced by these numerical methods is crucial in order to avoid erroneous conclusions.

This course follows along the lines of a course on mathematical physics and computational physics. It is designed to teach

- experimental techniques commonly used to measure current-voltage relationships of various samples and the associated errors,
- methods to analyze experimental data, extract interesting physics from it and relate to theory
- technical skills to operate and program sophisticated electronic instrumentation as well as handle vacuum ($\sim 3 \times 10^{-5}$ bar) and cryogenic systems (down to 77 K)

We begin by first describing the experimental techniques needed to accurately measure various regimes of the I-V phase space in Section II. Next, in Section III, we perform a series of measurements on devices with known I-V curves in order to familiarize the student with the experimental procedures and to build confidence in the instrument. After having checked various specifications of the instrument in Section III, we then proceed to characterize a conductor (Section IV), a superconductor (Section VI) and finally an insulator (Section V). We relegate precise operational procedures needed to operate the instrumentation to Section VII and a brief explanation of the software needed to perform the experiments to Section VIII.

II. UNDERSTANDING MEASUREMENT TECHNIQUES

Objectives

1. To understand the differences between 2- and 4-probe configurations, and identify the cases in which each should be used
2. To compare the resistances of a highly conductive sample measured using 2- and 4-probe configurations
3. To compare the resistances of a morderately resistive sample measured using 2- and 4-probe configurations

Requirements

1. `XPLORE` Source and Measurement Unit
2. `XPLORE` Sample Workbench

3. `XPLORE` Sample Puck with 2 cm copper sample attached

4. `XPLORE` Sample Puck with 1 km copper sample attached

Different materials such as conductors, insulators and superconductors (see Table I) occupy very different parts of the I-V phase space (shown in Fig. 2). The techniques and the instrumentation needed to perform the I-V measurements depend on where in this phase space the measurements are being performed.

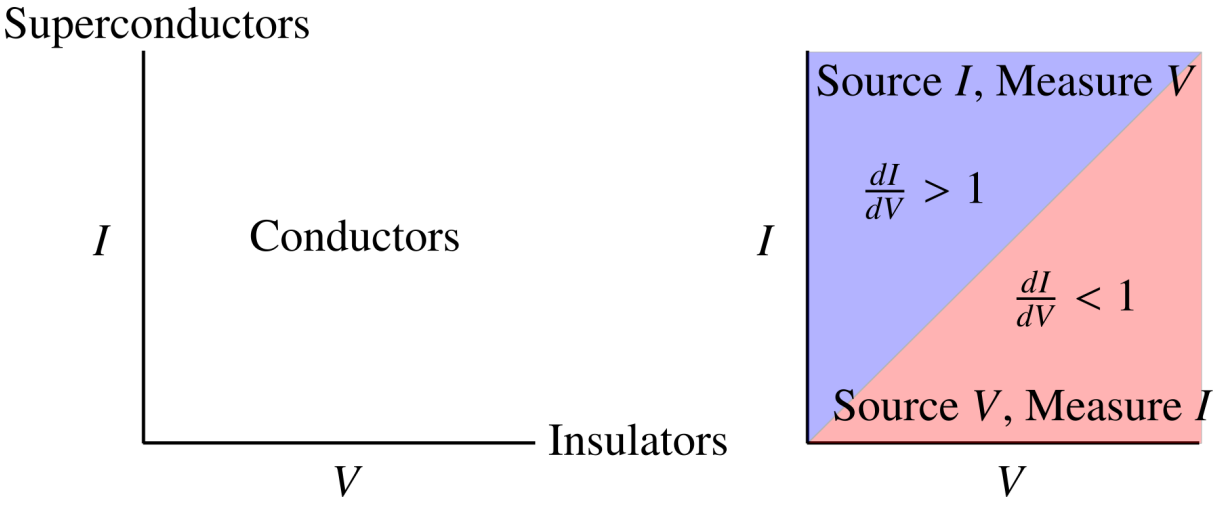


FIG. 2. (Left) Location of different materials in the I-V phase space, (right) Techniques that need to be used depending on parts of the phase space where measurements are being performed.

A. Choice of Source: Voltage or Current

The first step to measuring electrical transport properties of a sample is choosing a source to probe it with. A choice of the source is made depending on the location of measurements in the I-V space. Before we delve into the details of measurement techniques, let us review some fundamental concepts of electricity.

Voltage (V) is the difference in electric potential between two points and is defined as the work required to be done in order to move a unit charge from one point to the other in the presence of a static electric field. The SI unit of voltage is Volt. One Joule of work is required to be done in order to move one Coulomb of charge across a potential difference of one Volt.

The rate of flow of electric charge (Q) is electric current (I). Electric current arises due to the movement of electrons and/or ions in a medium.

$$I = \frac{dQ}{dt} \tag{2}$$

Electrical resistance of a material is a measure of the obstruction to the flow of electric current through it. Resistance (R) of an object is defined as the ratio of voltage across it (V) to current through it (I). $R = V/I$

In certain devices, positive electric charges flow from a higher to a lower potential energy (ie. conventional current flows from the positive to the negative terminal). This leads to the conversion of potential energy of these charges into kinetic energy. This kinetic energy is then 'lost' to the environment in the form of heat, light, etc. Such devices are known as passive devices, and they 'consume' power.

On the other hand, active devices are ones in which conventional current flows from the negative terminal to the positive terminal. These devices are sources of power.

What is a Voltage Source?

An ideal voltage source is a two terminal device that maintains a constant voltage difference across its two terminals, irrespective of the load connected across it. To maintain a constant voltage (V) across a load with resistance (R), a voltage source would need to supply a current ($I = V/R$) through it. As the value of load resistance decreases, the current required to maintain a voltage (V) through it increases. Due to various reasons like the resistivity of its component materials, or ionic properties of its chemicals, practical voltage sources have a finite *internal resistance*, which limits the current they can supply (current compliance). This sets a lower limit on the value of resistance across which the constant voltage (V) can be maintained.

The voltage source in the `XPLORE` Source-Meter Unit (SMU) can supply voltages upto 10 V with a 10 mA current compliance, and has a resolution better than 0.5 mV.

What is a Current Source?

An ideal current source is a device that can source a constant current through it, irrespective of the load connected across it. To maintain a constant current (I) through a load with resistance (R), a current source must be able to supply a voltage ($V = I \times R$) across it. As the load resistance (R) increases, a current source would need to supply an ever increasing voltage (V) across it to maintain the constant current (I). Practically, current sources have a finite supply voltage (called

the *compliance voltage*). *Compliance voltage* is the maximum value of voltage that the current source can possibly supply to a load. This sets an upper limit on the value of resistance through which the constant current can be maintained. Real world current sources have an high output resistance.

The current source in the `XPLORE` SMU is capable of sourcing currents upto 10 mA with a 10 V compliance voltage, and has a resolution better than 0.5 μA.

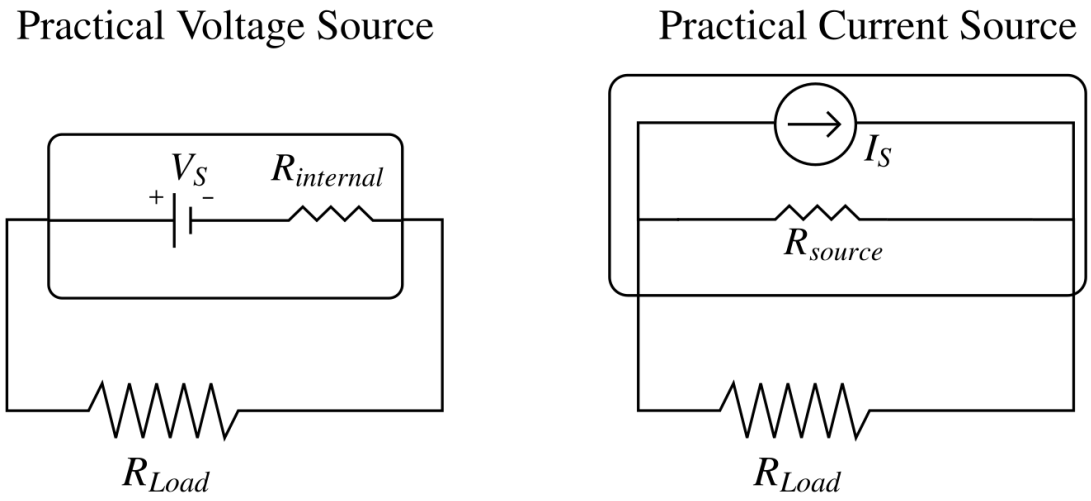


FIG. 3. Non-ideal Sources

Choosing the right source

Figure 4 shows various values of resistances each of which is probed with a 1 V voltage source and a 1 mA current source. The corresponding measured quantities for each excitation (calculated using $V = IR$) are shown in the adjacent columns in the table.

We observe that when a sample with a resistance lower than 10 Ω is excited using a 1 V source, the measured current becomes greater than 100 mA! Such high values of current pose several practical problems :

1. The maximum current that can be supplied by a voltage source is limited by its internal resistance. So the voltage source may not be able to maintain the set voltage across the device under test.

Resistance (Ohms)	Source Voltage	Mesaured Current	Source Current	Measured Voltage
...				
100 M	1 V	0.01μA	1 mA	100,000 V
10 M	1 V	0.1 μA	1 mA	10,000 V
1 M	1 V	1 μA	1 mA	1,000 V
100 K	1 V	10 μA	1 mA	100 V
10 K	1 V	100 μA	1 mA	10 V
1 K	1 V	1 mA	1 mA	1 V
100 E	1 V	10 mA	1 mA	0.1 V
10 E	1 V	100 mA	1 mA	0.01 V
1 E	1 V	1 A	1 mA	0.001 V
0.1 E	1 V	10 A	1 mA	0.0001 V
...				

Practically Challenging Measurements

Practically Feasible Measurements

FIG. 4. Choosing the Right Source[a]

[a] Regions shaded red denote values outside the measurement ranges of the `XPLORE` SMU

2. The device under test may get damaged on passing a large current through it. For example, YBCO (a high temperature superconductor) has a critical current that it can sustain. A current higher than this causes it to lose its superconductivity.

3. The connecting wires need to have high current capacity. This would mean having thicker wires, that are bulky and occupy more space. This does not work well with space-constrained environments

On the other end, for similar reasons it is not feasible to use a current source excitation to probe a sample with resistance greater than 10 $K\Omega$

1. A current source has a finite supply voltage, called its compliance voltage.

2. Measurement of high voltages requires components that are rated for such voltages. These

increase the cost of the equipment.

3. High power ($V \times I$) dissipated across the device under test leads to heating and subsequent damage of the device.

The observant reader will notice that resistances ranging from 100 Ω to 10 $K\Omega$ can be probed with either source. The table in figure 4 is given as a general rule of thumb, and may be deviated from in special cases.

B. Contact geometry: Two-Probe vs Four-Probe

Electrical contacts to the sample are arguably the single most important consideration in measuring electrical transport properties. Electronic transport measurements are made using metal wires/probes attached to the device under test. When a metal is used to make electrical contact to a sample, there are several issues that may arise :

1. *Contact Resistance* arises at the interface of these two materials due to dirt or oxides on the surfaces. If not properly handled, these resistances contribute to the value of the measured resistance, which provides an inflated measurement of resistance.

2. Formation of non-linear *Schottky barrier* with rectifying characteristics. A Schottky barrier arises due to the mismatch of Fermi levels at metal-semiconductor junctions. These junctions have non-linear characteristics of their own, which make probing the sample useless by blocking the flow of charge between the sample and the external circuitry. The primary characteristic of a Schottky barrier is the Schottky barrier height (Φ_B), which depends on the combination of metal and semiconductor. A contact that conducts in both directions without rectification is known as an Ohmic contact, which may form when the Schottky barrier height is very low.

In order to accurately determine the I-V characteristics of a sample, low-resistance, stable Ohmic contacts to samples are critical. Achieving Ohmic contacts in metal-semiconductor junctions are possible but require specific techniques like deposition of thin metal films layered in a specific order, followed by post-processing (annealing) to alter the semiconductormetal bond. On the other hand, in metal-metal junctions, Ohmic contacts can be easily made by techniques like soldering or welding, but care must be taken to ensure direct contact of metals, without intermediate layers of insulators or oxides.

In the forthcoming sections, we consider Ohmic contacts to the sample with a finite contact resistance, and focus on techniques to eliminate this contact resistances from I-V measurements. We start with a standard "2-probe" configuration (where the sample is contacted by 2 measurement probes), and discuss systematic modifications required to solve the aforementioned problem.

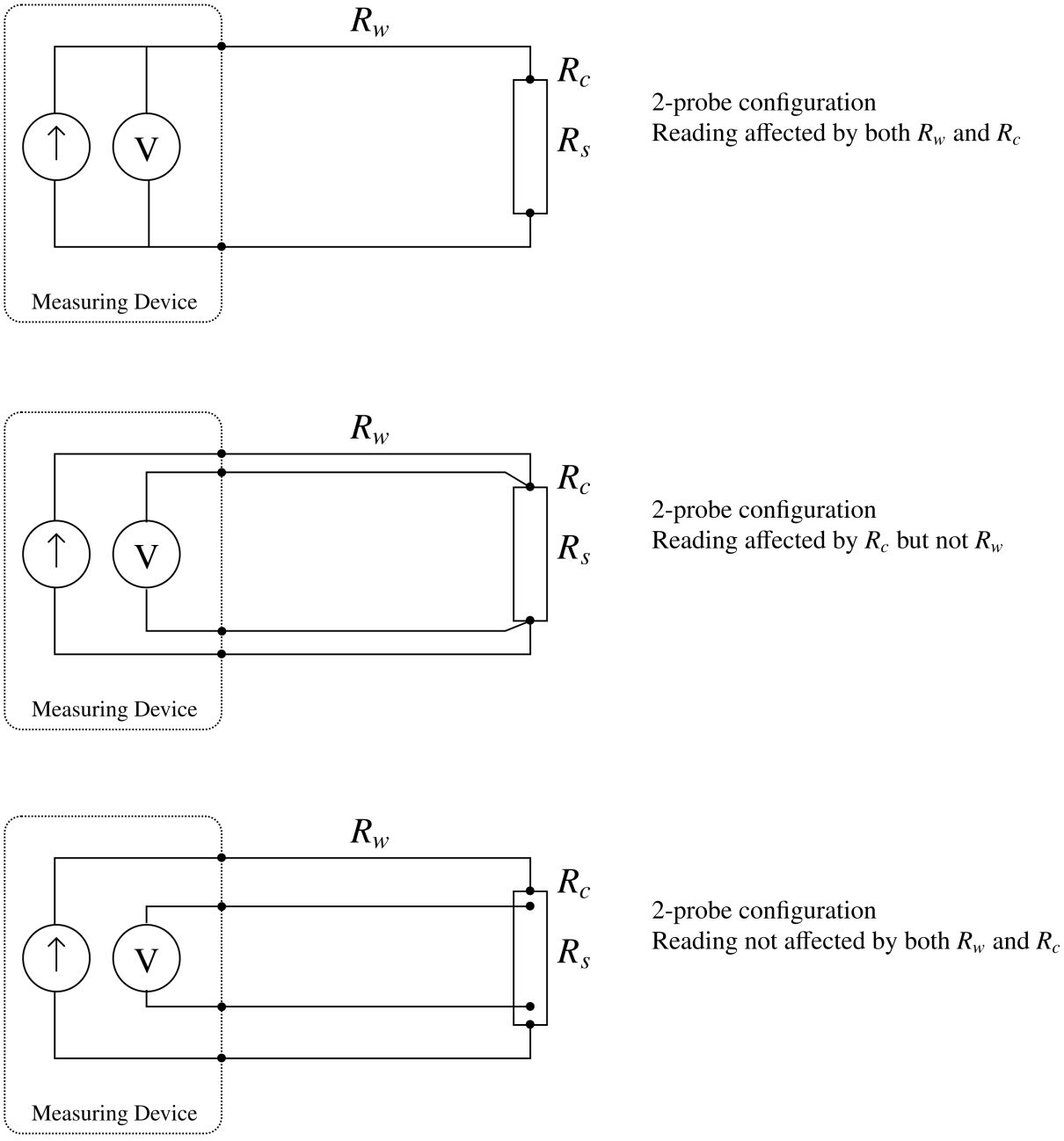


FIG. 5. Electronic Transport Property Measurement Configurations

Figure 5a shows the schematic for the 2-probe measurement scheme. The current source supplies

a constant current (I) to the sample. A voltmeter in parallel to the current source measures the voltage (V). The resistance (R_s) of the sample is calculated as $R_s = V/I$. Note that only two wires (and 2 contacts to the sample) are required to perform electrical transport measurements here. Now let us discuss why this approach can be inaccurate.

If we probe the sample in this configuration, the current supplied by the current source will also cause a voltage drop across the wires (R_w) and the contact resistances (R_c), which is being measured by the voltmeter because it is connected across all of these (sample + contacts + wires). This causes erroneous measurement of the sample's resistance as the measured R is actually $R_s + R_c + R_w$ and not just R_s. In fact, wire and contact resistances will dominate when samples with very low resistance are probed this way. Therefore, this measurement technique is inaccurate for such samples.

If we move the probes of the voltmeter closer to the sample (as in figure 5b), we have removed the resistance of the wires (R_w) from the equation. The resistance of the contacts (R_c) still remains as the voltmeter is connected across the contacts and the samples. For samples with a resistance comparable to the contact resistances, the measurement will not be accurate.

Moving the voltmeter probes even closer, slightly inward of the current source contacts (figure 5c) we have what is known as the linear 4-probe configuration. The voltmeter probes are connected only across the sample in this case. Note that for resistivity calculations (that require the geometry of the sample to be known), the length of the sample within the voltmeter probes is taken as the sample length.

Let us illustrate 2- and 4-probe measurements with some simple experiments. We take 2 copper wire samples (of diameter 0.1219 mm, SWG 40) with different lengths (1 cm and 1 km[1]). We measure their resistances and compare our measured values to their expected calculated resistances.

Moderate to High Resistance Samples

The expected resistance of the 1 km long copper wire can be calculated as :

[1] A direct measurement of a one kilometer sample is impractical. Given the diameter of the wire, and the density of copper, we calculate and measure out the weight of copper wire (neglecting the weight of the insulating layer of enamel) that would be needed for the wire to be 1 km in length.

$$
\begin{aligned}
R &= \rho \frac{l}{A} \\
&= 1.68 \times 10^{-8} \times \frac{1000}{\pi \times (0.0001219)^2} \\
&= 1439.544\ \Omega
\end{aligned} \tag{3}
$$

We now measure the resistance of this wire in 2-probe configuration.

1. Mount the copper wire sample onto the `sample puck`, and mount the `sample puck` on the `sample workbench`.

2. Connect the banana plugs from the `XPLORE` Source and Measurement Unit to the `sample workbench` as shown in figure 6.

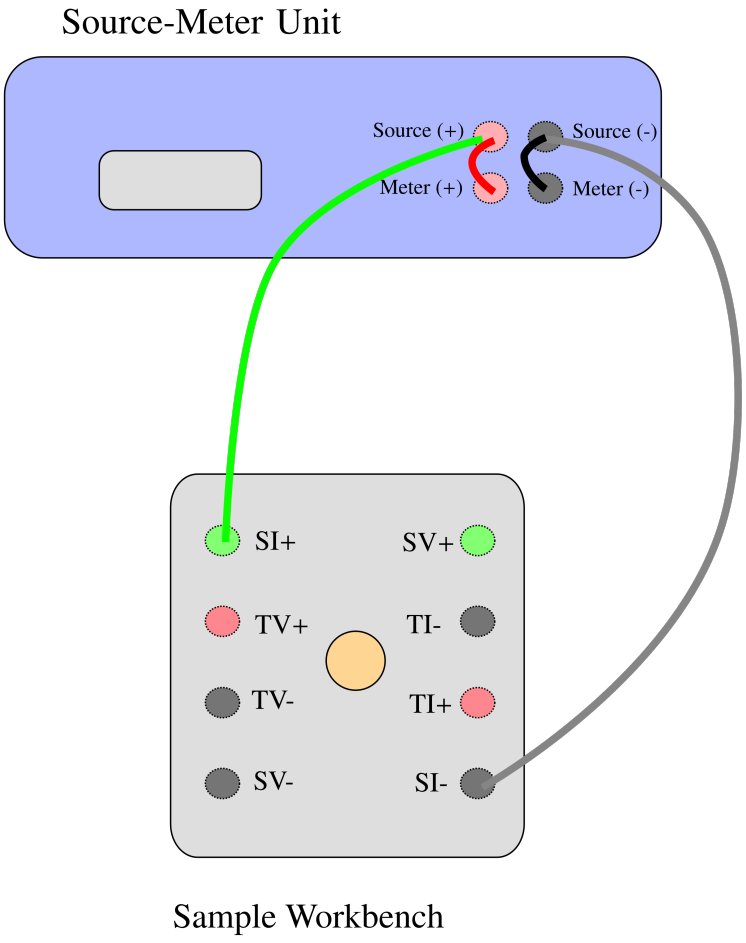


FIG. 6. Setting up resistance measurement of a copper wire in 2 probe configuration

3. In `Qrius`, click on `Modules Manager` → `Source and Measurement Unit`. On the top of the `Source and Measurement Unit` window, click on the `Settings` → `Source Parameters`.

4. Make sure that the `source mode` is set to `Constant Current`, and set the value of current to $5,000\ \mu A$. Click on `File` → `Apply` to save the settings.

5. On the left side of the `Source and Measurement Unit` window, make sure that the `Ammeter` reports a value close to the set value of current. Note down the `Measured Voltage`.

6. Using the values sourced current (I) and measured voltage (V), calculate the resistance (R) of the sample ($R = V/I$).

7. Compare this measured value of resistance to the value calculated in equation 3 by calculating the absolute and percentage errors.

8. Reset the value of source current to 0 μA from the `Source Parameters` window. Click on `File` → `Apply` to save the settings.

Conclusion

1. Calculated Resistance : _____ Ω

2. Measured Resistance : _____ Ω

3. Absolute Error : _____

4. Percentage Error : _____

Low to Moderate Resistance Samples

We repeat the above exercise for the shorter copper wire sample. Again, the expected resistance is calculated as :

$$\begin{aligned} R &= \rho \frac{l}{A} \\ &= 1.68 * 10^{-8} \times \frac{0.01}{\pi \times (0.0001930)^2} \\ &= 0.1439\ \Omega \end{aligned} \tag{4}$$

Resistance measurements on the sample are performed as follows :

1. Repeat the resistance measurement procedure as performed with the longer wire, keeping the source current value the same as before. Note the value of measured resistance.

2. Compare this measured value of resistance, with the value calculated in equation 4.

3. In the `Source Parameters` window, reset the value of current to 0 μA and click on `File` $\rightarrow$ `Apply`

Conclusion

1. Calculated Resistance : _____ Ω

2. Measured Resistance : _____ Ω

3. Absolute Error : _____

4. Percentage Error : _____

We see that percentage error is much higher in the measurement of the shorter wire, as the contact resistances and resistance of the probe leads are now comparable (or even higher) than the actual resistance of the sample. The configuration we have used up to now is called the 2-probe configuration. Let us repeat our measurements of the shorter copper wire with what is known as the 4-probe configuration.

1. Disconnect the voltage measurement plugs (V+ and V-) from the source plugs, and connect them to the sample workbench (I+ and I- respectively) as shown in figure 7.

2. Repeat the resistance measurement procedure (supply a constant current of 5000 μA and measure the voltage as before).

3. Calculate the measured resistance, and compare it to the value of expected resistance (ref. equation 4) by calculating the absolute and percentage errors.

Conclusion

1. Calculated Resistance : _____ Ω

2. Measured Resistance : _____ Ω

3. Absolute Error : _____

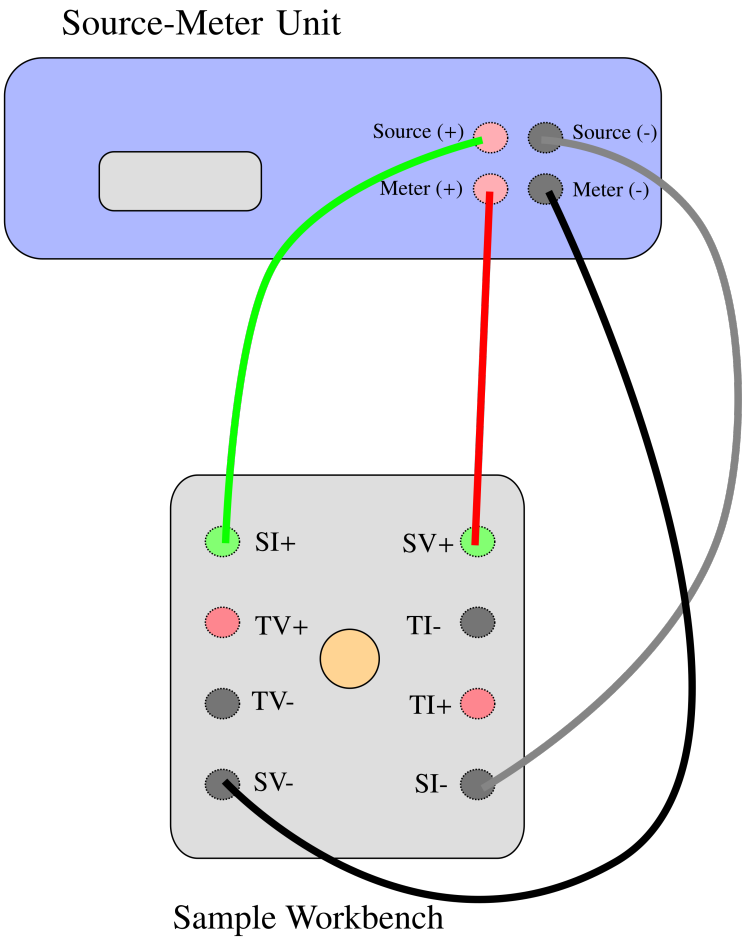


FIG. 7. Setting up resistance measurement of a copper wire in 4 probe configuration

4. Percentage Error : _____

Which measurement technique seems suitable for resistance measurement of the shorter copper wire?

Similarly, we may also perform a 4-probe measurement on the longer copper sample and check that there is no degradation in the accuracy of the measurement when compared to the shorter sample.

Thus we conclude that to accurately measure electronic transport properties of samples with low resistances, it is essential to use a 4-probe configuration. For samples with high resistances, 2-probe configuration results in sufficiently accurate measurements.

III. WARM-UP EXPERIMENTS : GETTING TO KNOW YOUR INSTRUMENT

We begin the first set of experiments by developing familiarity with the operational procedures, understanding the various measurement techniques and cross-checking specifications of the instrumentation.

A. Characterizing Errors in Temperature Measurement

Objectives

1. To understand the operation of vacuum and cryogenic systems
2. To cross-check the temperature reading on the `XPLORE` Physical Quantities Measurement System
3. To determine the experimental error in sample temperature

Requirements

1. `XPLORE` Source and Measurement Unit
2. `XPLORE` Temperature Controller
3. `XPLORE` Low temperature cryostat with evacuation and flushing
4. `XPLORE` Variable Temperature Insert
5. `XPLORE` Sample Puck with PT100 sample attached

A good experimentalist does not blindly trust the equipment, but rather verifies the fidelity of each of its functions by subjecting it to intelligently designed unit test experiments[2]. In this section, we design experiments to probe and optimize the accuracy of temperature measurement in the `XPLORE` Physical Quantities Measurement System.

We employ a Resistance Temperature Detector (RTD) in our endeavour to cross-check the *Sample Temperature* displayed by the `TCON`. A `PT100` is a high precision RTD which consists of a

[2] In his famous commencement address at Caltech, Feynman posited how science should be practiced - "I would like to add something that's not essential to the science, but something I kind of believe, which is that you should not fool the laymen when you're talking as a scientist... I'm talking about a specific, extra type of integrity that is not lying, but bending over backwards to show how you're maybe wrong, [an integrity] that you ought to have when acting as a scientist. And this is our responsibility as scientists, certainly to other scientists, and I think to laymen." We strive to keep this spirit alive in all our work.

Platinum thin film or a Platinum wire wrapped around a ceramic/glass core. We solder the `PT100` sample on the Sample Puck in 4 probe configuration, ensuring good thermal contact between the sample body and the copper block using thermally conductive vacuum (Apiezon) grease. We will then source a constant current of 1 mA through it and subject the sample to various temperature set-points in the range 77-450K, while measuring the voltage across its terminals.

Why choose an RTD?

RTDs have a linear output over a wide range of temperature and provide greater accuracy than other temperature sensors like thermocouples, Negative Temperature Coefficient (NTC) thermistor and semiconductor based sensors. Moreover, the measurement of temperature using an RTD requires just a constant current source and a voltmeter, both of which are available in the `XPLORE` PQMS. We encourage the user to perform the similar experiments with other types of temperature sensors.

The experimental procedure is described below :

1. The `sample puck` mounted with a `PT100` in 4-probe configuration is screwed onto the `variable temperature insert` (ref. section VII A 1).

2. The insert is loaded into the cryostat (ref. section VII E).

3. The sample and heater chambers are flushed with helium exchange gas as discussed in sections VII C 2 and VII C 1

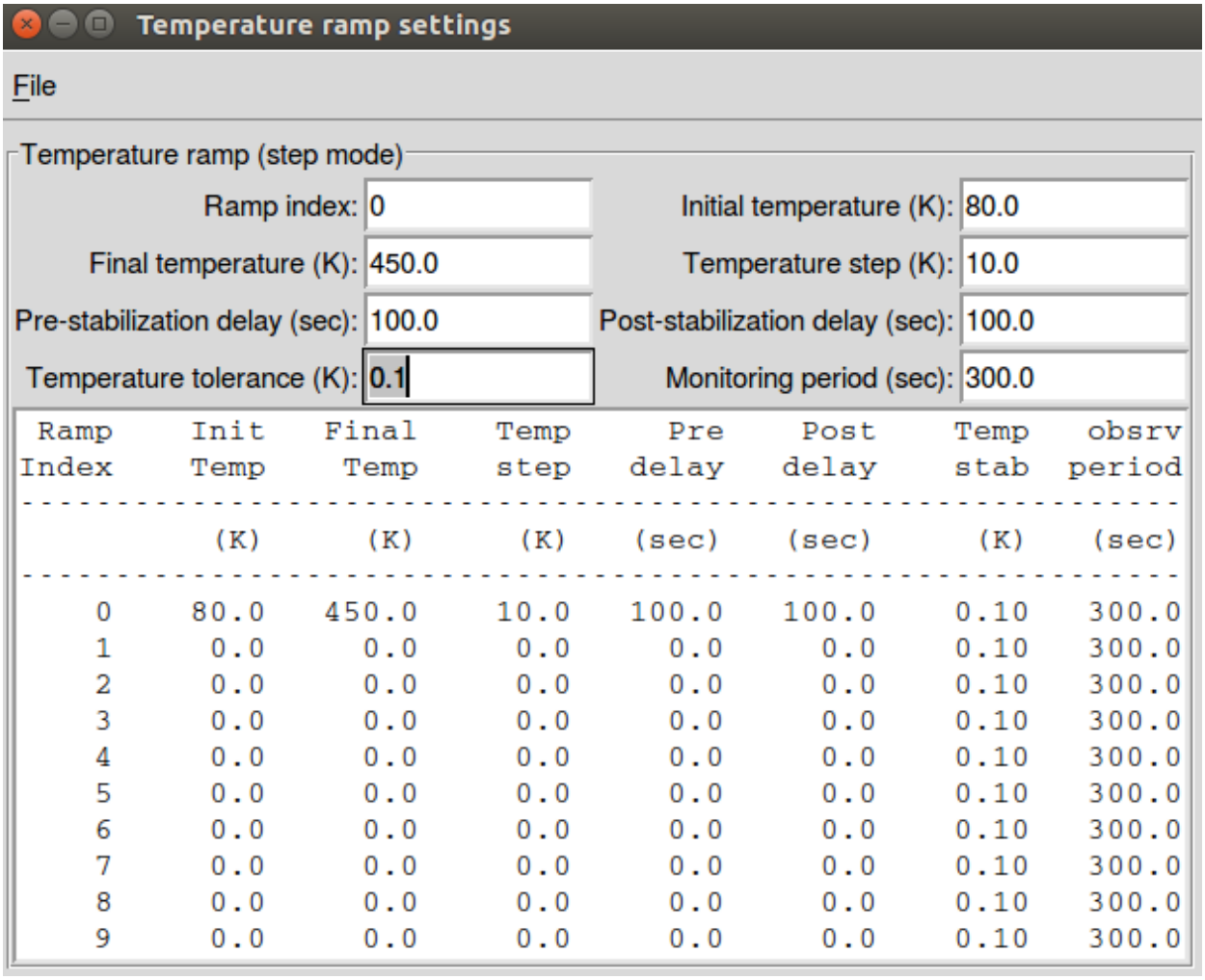
4. Liquid nitrogen is poured into the cryostat to cool the sample down (ref. section VII D).

5. To configure `Qrius` to measure resistance vs temperature (stepped ramp), navigate to `Measurement Mode` → `Electrical DC Conductivity Mode` from `Qrius`'s main menu. In the Electrical DC Conductivity window, select the `Run Mode` as `R-T (Stepped Ramp)`.

6. To configure resistance measurement settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `IV Source and Measurement Unit` → `Resistance Measurement Settings`. Set the Current Limit to 1 mA, Voltage Limit to 5000 mV, and Power Limit to 100 mW. Click on `File` → `Done` to save the settings.

7. To configure the temperature stepped ramp settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `Temperature Controller` → `Step`

`Ramp Settings`. Configure the step ramp settings as shown in figure 8. Click on `File` → `Apply` to save the settings.



Ramp Index	Init Temp (K)	Final Temp (K)	Temp step (K)	Pre delay (sec)	Post delay (sec)	Temp stab (K)	obsrv period (sec)
0	80.0	450.0	10.0	100.0	100.0	0.10	300.0
1	0.0	0.0	0.0	0.0	0.0	0.10	300.0
2	0.0	0.0	0.0	0.0	0.0	0.10	300.0
3	0.0	0.0	0.0	0.0	0.0	0.10	300.0
4	0.0	0.0	0.0	0.0	0.0	0.10	300.0
5	0.0	0.0	0.0	0.0	0.0	0.10	300.0
6	0.0	0.0	0.0	0.0	0.0	0.10	300.0
7	0.0	0.0	0.0	0.0	0.0	0.10	300.0
8	0.0	0.0	0.0	0.0	0.0	0.10	300.0
9	0.0	0.0	0.0	0.0	0.0	0.10	300.0

FIG. 8. Temperature Stepped Ramp Settings [a]

[a] Parameter values may be changed as per the goal/time constraint of the experiment. For a detailed explanation of the various parameters in the temperature stepped ramp settings, refer to section VIII C 1

8. Start the `R-T` run by clicking on the `Start` button on the `Electrical DC Conductivity` window. Wait for the acquisition to end. This may take a while depending on the temperature range and number of temperature setpoints at which data is being acquired.

9. Once the acquisition has ended, the measured resistance of the `PT100` for each temperature set-point can be converted into a corresponding temperature reading using the `PT100`'s calibration table. The measured temperature is plotted with respect to the `TCON`'s' sample

temperature in figure 9.

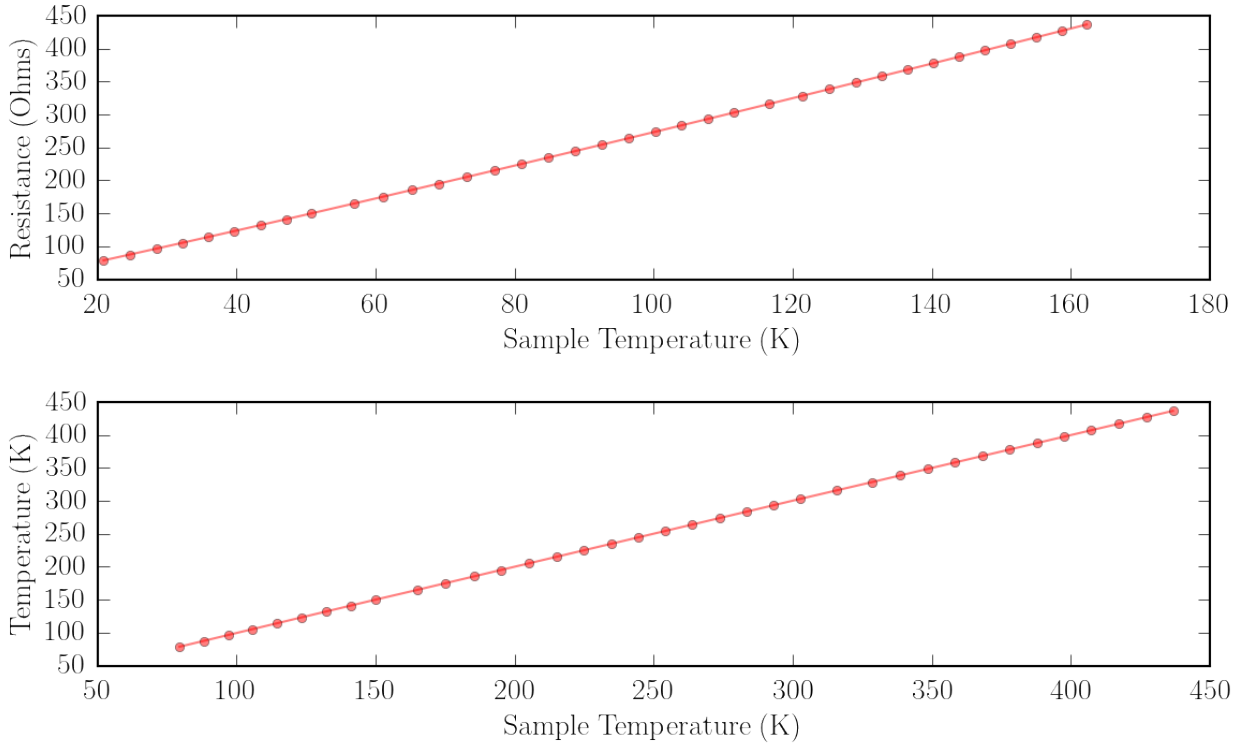


FIG. 9. (1) Measured Resistance and (2) Measured Temperature vs `TCON`'s Sample Temperature Reading

10. The value of temperature measurements obtained this way are compared to the corresponding `TCON`'s Sample Temperature readout to determine the error at each temperature setpoint (ref. 10).

Can we trust the RTD calibration table?

In the above experiment we have used the calibration table provided by the `PT100` manufacturer to convert its measured resistance into a temperature reading. But a good experimentalist has no reason to trust the calibration table. A first cross-check for the calibration table would be to measure the resistance of the `PT100` sensor when dipped in ice-water and boiling water. Pure water boils at 100° *Celcius* at sea level, but at 93.4^o C at 2000 meters altitude. In the presence of nucleating substances, water freezes at 0^oC. Further checks can be performed with solutions of Ethanol and Water mixed in various ratios. We leave it up to the good experimentalist to perform these experiments.

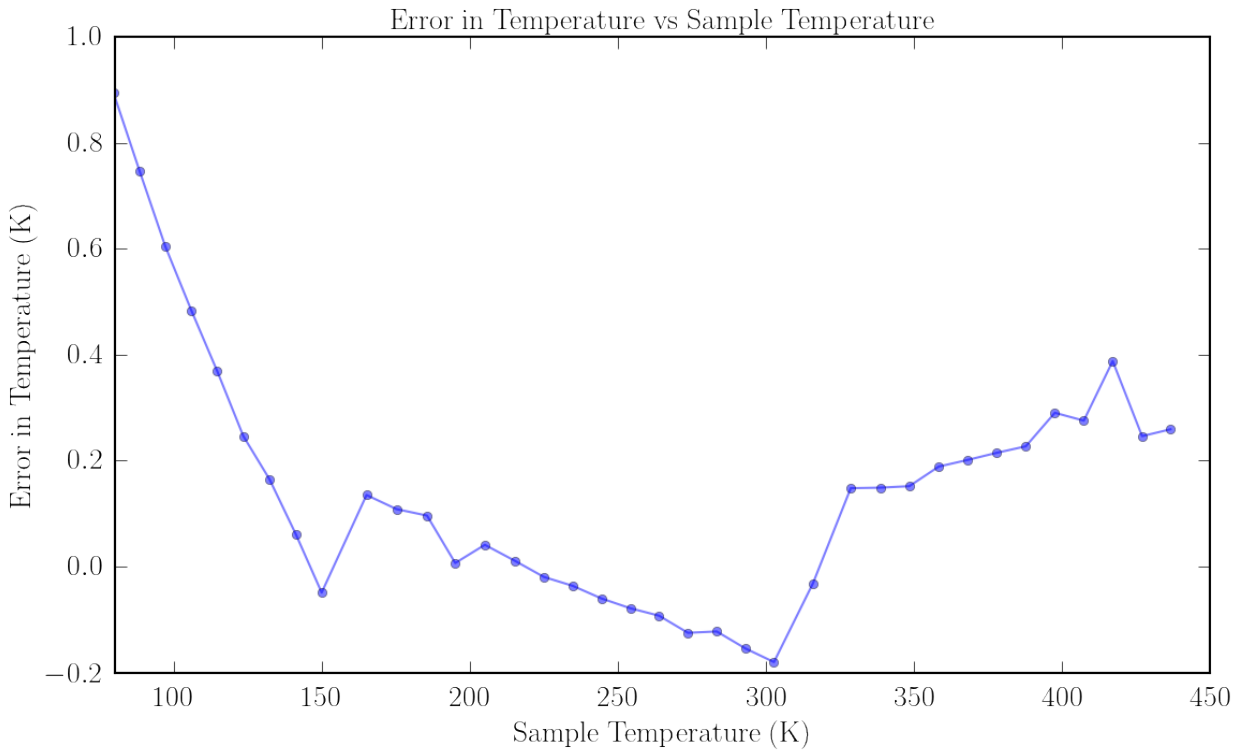


FIG. 10. Error in Temperature vs `TCON`'s Sample Temperature Reading

B. Intelligent Source Switching and the Characterization of Non-linear Devices

Objectives

1. To understand the selection of source based on the characteristics of sample under test

2. To measure the current-voltage characteristics of a non-linear device at various temperatures

Requirements

1. `XPLORE` Source and Measurement Unit

2. `XPLORE` Temperature Controller

3. `XPLORE` Low temperature cryostat with evacuation and flushing

4. `XPLORE` Variable Temperature Insert

5. `XPLORE` Sample Puck with a zener diode attached

As discussed in section II A, voltage sources are more suited for transport measurements of highly resistive samples, whereas a current source is needed to probe samples with low resistance. To illustrate this better, let us look at the I-V characteristics of a non-linear device, which has regions of both, high and low resistance.

Zener Diodes are highly doped p-n junctions, that allow current to flow not only from its anode to cathode, but also in the reverse direction when the Zener voltage is reached.

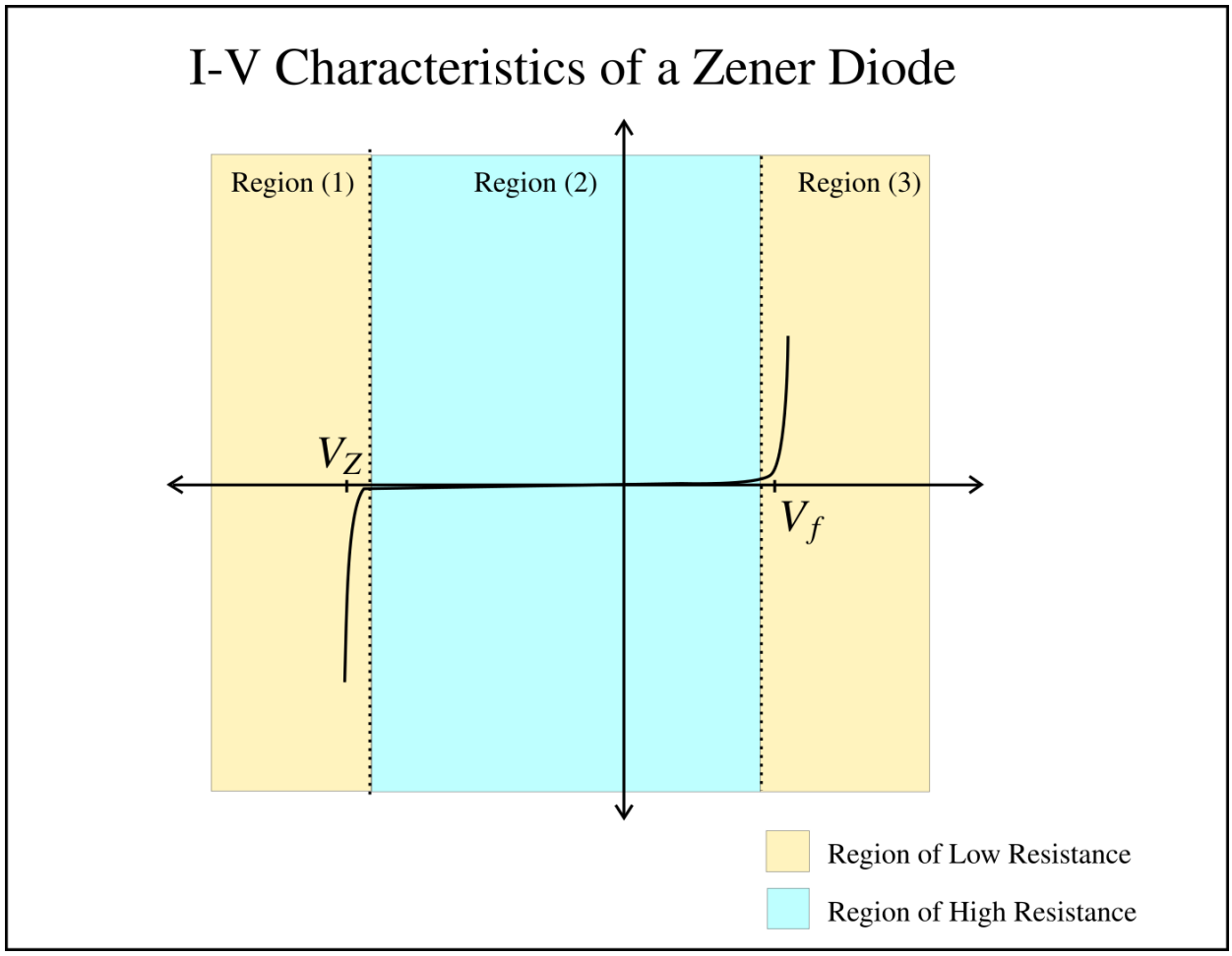


FIG. 11. Typical I-V characteristics of a Zener Diode - Resistance of the same device is varies drastically in different parts of the I-V space

Consider the I-V characteristics of a Zener diode shown in figure 11. In region (2), the Zener diode behaves as a device with very high resistance. On the other hand, in regions (1) and (3), the same Zener diode shows a very low resistance.

If we were to choose a voltage source to probe the transport properties of the Zener diode over its entire I-V space, we would end up with useful measurements only in region (2), where the

sample shows a high resistance. However, if we were to choose a current source to probe the Zener diode, we would end up with useful measurements in regions (1) and (3), but not in region (2). This change in nature of the device from highly resistive to highly conductive poses a problem in choosing a single source that can appropriately probe its properties.

In `Qrius`, the I-V measurement mode can automatically switch between a voltage source and current source, depending on the resistance of the device under test at that particular point in its I-V space. Let us see this in action with a Zener diode (`BZX79-B6V2`).

1. The sample puck mounted with a `Zener diode` in 4-probe configuration is mounted into the `variable temperature insert` (ref. section VII A 1).

2. The insert is loaded into the cryostat (ref. section VII E).

3. The sample and heater chambers are flushed with helium exchange gas as discussed in sections VII C 2 and VII C 1.

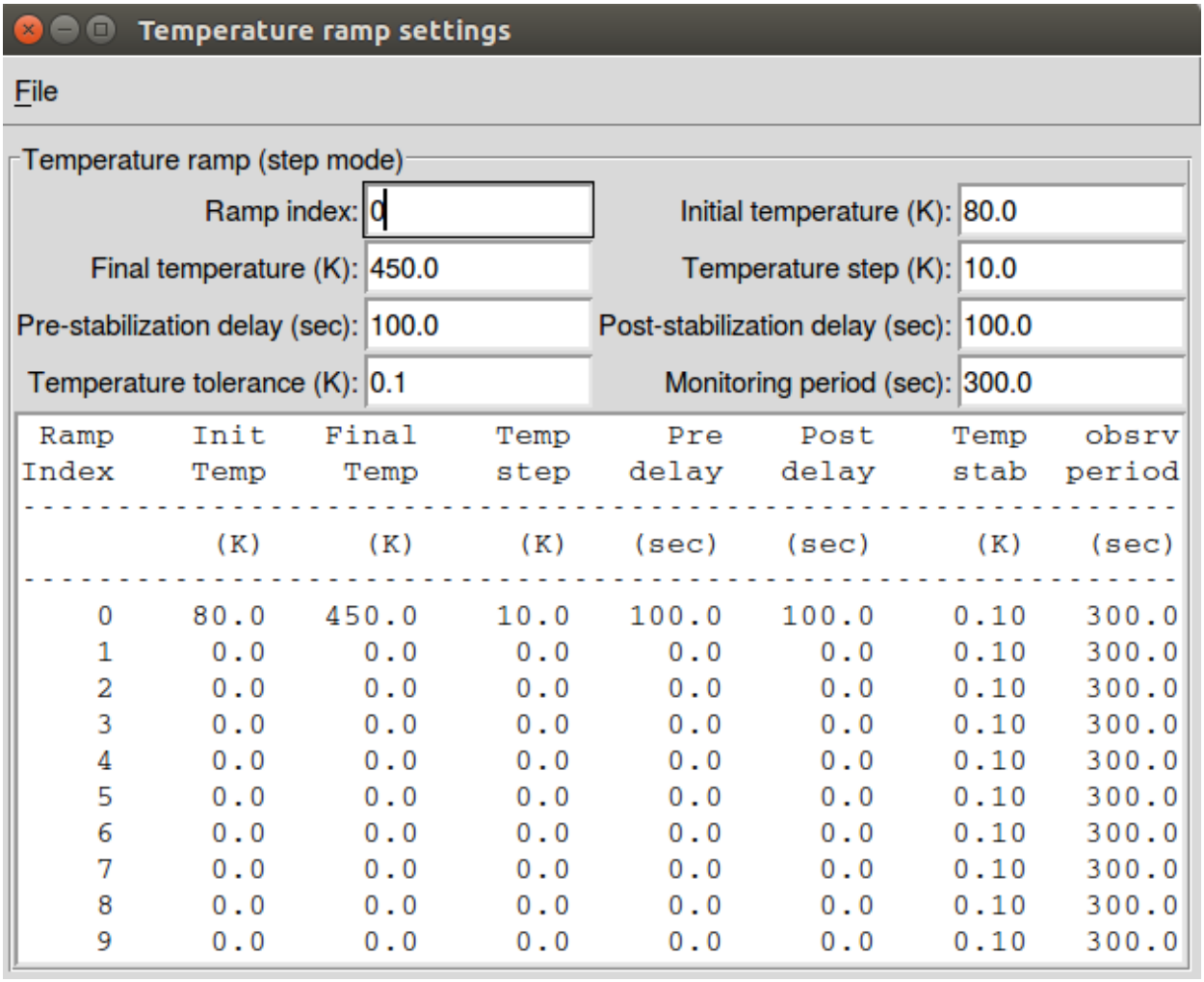
4. If I-V below room temperature is required, liquid nitrogen is poured into the cryostat to bring the sample temperature down (ref. section VII D).

5. To configure `Qrius` to measure I-V vs Temperature (Stepped Ramp), go to `Measurement Mode` → `Electrical DC Conductivity`. In the `Electrical DC Conductivity` window, select the `Run Mode` as `I-V (Stepped Ramp)`.

6. In the `Electrical DC Conductivity` window, click on `Settings` → `IV Source and Measurement Unit` → `I-V Measurement Settings`. Set the current, voltage and power limits to 10000 μA, 10000 mV, and 100 mW respectively. Set the current and voltage step sizes to 500 μA and 100 mV. Make sure `Bipolar` is set to `Yes`. Click on `File` → `Done` to save these settings. An explanation of the I-V measurements settings have been provided in section VIII C 2

7. In the `Electrical DC Conductivity` window, click on `Settings` → `Temperature Controller` → `Step Ramp Settings`. Configure the step ramp settings as shown in figure 12. Click on `File` → `Apply`.

8. Start the I-V vs T run by clicking on the `Start` button on the `Electrical DC Conductivity` window. Wait for the acquisition to end. This may take a while depending on the current/voltage limits and step sizes, and the number of temperature setpoints at which data is being acquired.



Ramp Index	Init Temp (K)	Final Temp (K)	Temp step (K)	Pre delay (sec)	Post delay (sec)	Temp stab (K)	obsrv period (sec)
0	80.0	450.0	10.0	100.0	100.0	0.10	300.0
1	0.0	0.0	0.0	0.0	0.0	0.10	300.0
2	0.0	0.0	0.0	0.0	0.0	0.10	300.0
3	0.0	0.0	0.0	0.0	0.0	0.10	300.0
4	0.0	0.0	0.0	0.0	0.0	0.10	300.0
5	0.0	0.0	0.0	0.0	0.0	0.10	300.0
6	0.0	0.0	0.0	0.0	0.0	0.10	300.0
7	0.0	0.0	0.0	0.0	0.0	0.10	300.0
8	0.0	0.0	0.0	0.0	0.0	0.10	300.0
9	0.0	0.0	0.0	0.0	0.0	0.10	300.0

FIG. 12. Temperature Stepped Ramp Settings [a]

[a] For a detailed explanation of the various parameters in the temperature stepped ramp settings, refer to section VIII C 1

9. Once the acquisition has ended, the IV data for various temperatures is plotted in semilog scale.

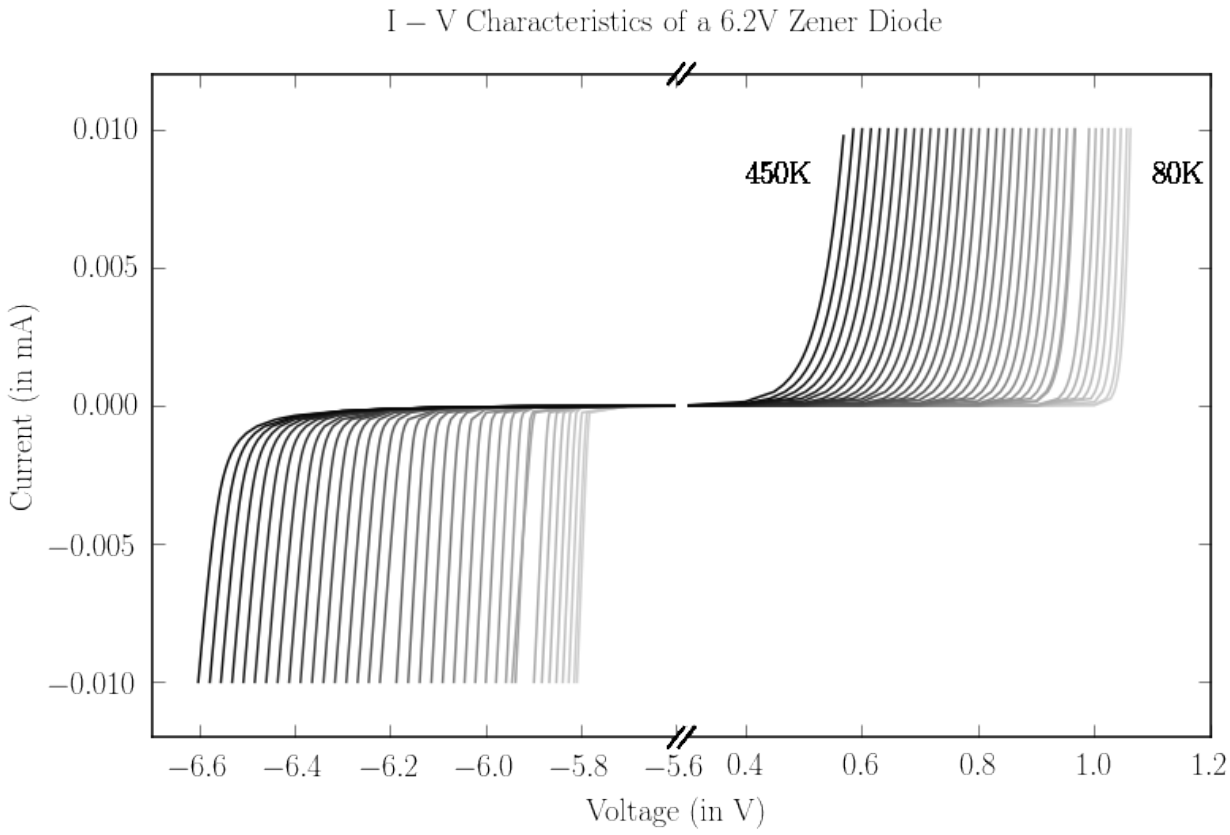


FIG. 13. I-V Characteristics of a Zener Diode at Various Temperatures

C. Characterizing Uncertainity in Resistance Measurements

Objectives

1. To make multiple measurements of the resistance of a precision resistor
2. To analyse the histogram of the resistance measurements to obtain the mean and error in measurement

Requirements

1. `XPLORE` Source and Measurement Unit
2. `XPLORE` Temperature Controller
3. `XPLORE` Low temperature cryostat with evacuation and flushing
4. `XPLORE` Variable Temperature Insert

5. `XPLORE` Sample Puck with a precision resistor attached

A resistor is a two terminal device that restricts the flow of electric current through it. Let us first define the characteristics of a resistor in terms of its voltage and current.

For a resistor,

$$V \propto I \tag{5}$$

where V is the voltage difference across the resistor's terminals, and I is the current flowing through the resistor.

The proportionality constant involved in this relationship is known as *Resistance*, denoted by R.

$$V = RI \tag{6}$$

A precision resistor has a resistance that lies very close to its nominal value (within a specified tolerance). They have a low temperature coefficient of resistance and are designed to maintain their resistance within a tolerance band over a specified range of temperature.

The relevant characteristics for a precision resistor `PTF561K0000BZEB` have been summarized in table II.

Property	Value
Resistance	1 $k\Omega$
Tolerance	$\pm 0.1\%$
Power	0.125 W
Temperature Coefficient	$5 ppm/^{\circ}C$
Operating Temperature	-55 *to* $150^{\circ}C$

TABLE II. Summary of Specifications for `PTF561K0000BZEB`

Let us measure the resistance of a precision resistor at room temperature (293.15 K).

1. The sample puck with the precision resistor (`PTF561K0000BZEB`) is mounted onto the `variable temperature insert` (section VII A 1), and the insert is loaded into the cryostat (section VII E).

2. The cryostat's sample and heater chambers are then flushed with helium (sections VII C 1 and VII C 2).

3. Next, liquid nitrogen is poured (ref. section VII D) to cool down the sample temperature to below 293.15 K (if the ambient temperature is above 293.15 K).

4. To configure the `TCON` to run in `Isothermal` mode, click on `Settings` → `Isothermal Settings` from the top of the `Temperature Controller` window. Set the `Heater Setpoint` to 293.15 K. Click on `File` → `Apply` to save the settings.

5. Click on `Start` in the `Temperature Controller` window. Wait for the sample temperature to stabilize to 293.15 K.

6. To configure the `SMU` to run in R-Time mode, click on `Settings` → `Resistance Measurement Settings` from the top of the `XPLORE Source and Measurement Unit` window. Set the current, voltage and power limits to 10000 μA, 5000mV and 100 mW respectively. Make sure that `Bipolar` is set to `Yes`. Click on `File` → `Done` to save the settings.

7. Click on `Start` on the `XPLORE Source and Measurement Unit` window. Let the acquisition run for about 30 minutes (1800 seconds).

8. Once sufficient number of datapoints have been gathered for analysis, click on `Finish` to stop the acquisition.

9. A histogram of all the measured data points is plotted by clicking on the 'histogram toggle button' under the plot.

10. The histogram is fitted with a Gaussian distribution (as shown in fig. 15). The spread (range), mean ($\bar{R}$) and standard deviation (σ_R) of this Gaussian fit are noted.

Conclusion

1. Calculated Resistance : _____ Ω

2. Measured Mean Resistance : _____ Ω

3. Range of Resistance : _____ Ω

4. Standard Deviation from Mean : _____ Ω

5. Absolute Error : _____

6. Percentage Error : _____

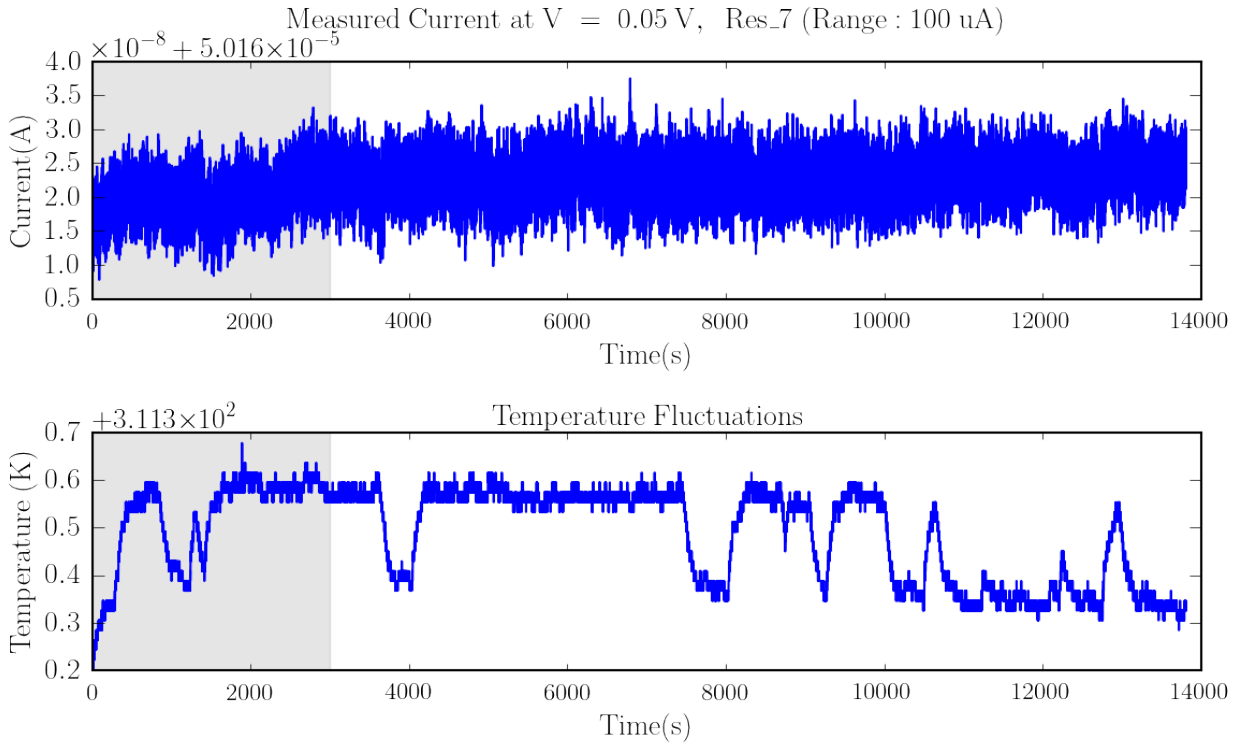


FIG. 14. Resistance-Time Series

Thus, in scientific text, the resistance of this resistor would be specified completely as $R = \bar{R} \pm \sigma_R$.

Extra credit question: Why do the measurements fall on a Gaussian? The student can repeat the experiment with different values of filter length (ref. `Acquisition Settings` in section VIII C 2) to check that the histogram converges more and more about the mean as the filter length increases (as determined by the central limit theorem).

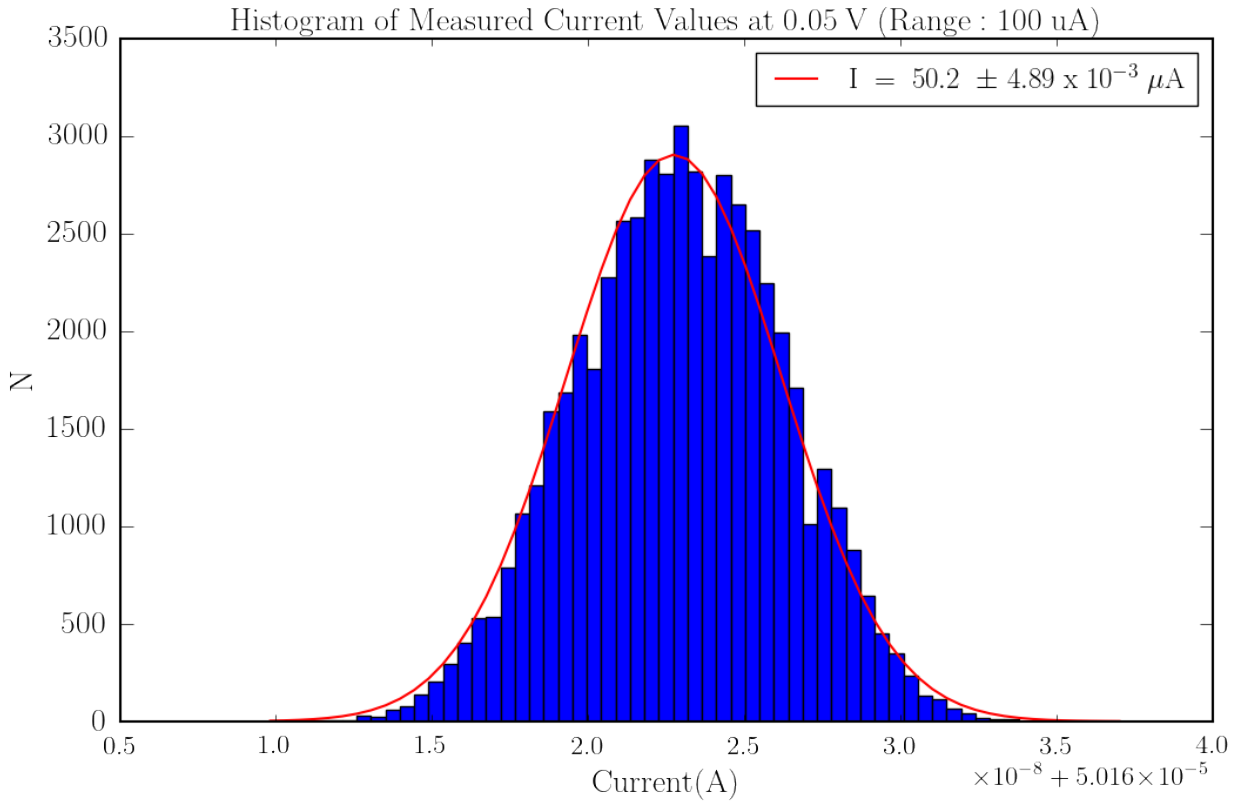


FIG. 15. Resistance Histogram

IV. CONDUCTORS

A. Measuring the Temperature Coefficient of Resistance of Copper

Objectives

1. To observe the variation of resistance with temperature for a cylindrical copper wire

2. To plot the variation of resistivity with temperature for copper

3. To calculate the temperature coefficient of resistance for copper at room temperature

Requirements

1. `XPLORE` Source and Measurement Unit

2. `XPLORE` Temperature Controller

3. `XPLORE` Low temperature cryostat with evacuation and flushing

4. `XPLORE` Variable Temperature Insert

5. `XPLORE` Sample Puck with a copper sample attached

Copper is one of the most widely used metals today. In research and development labs, its applications include device contacts, thin films, printed circuit boards and electrical equipment such as wiring, solenoids, heater coils and motor windings. Its wide usage in precision research equipment and devices makes copper an ideal candidate for electronic transport measurements.

In this experiment, we use commercially available 40 SWG (Standard Wire Gauge) copper wire samples of different lengths (2 *cm* and 5 *cm*), and determine the resistivity of Copper (at room temperature - 293.15 K) by measuring the current-voltage characteristics of the samples. We compare the value of resistivity obtained, to that provided by the IACS (International Annealed Copper Standard). We then go on to measure the current-voltage characteristics of the samples over a range of temperatures (200-400K) and see how the resistivity varies as a function of temperature. From these measurements we extract the temperature coefficient of resistivity for copper and compare it to standard published values for the same. Finally, we add error bars to the data based on our discussions in section III C.

The copper wire samples that we have chosen for our measurements have resistances comparable to those of the connecting wires, which make it necessary for us to use the 4-probe technique in order to eliminate these extraneous contributions.

1. The sample puck with the 2 *cm* copper wire is mounted onto the `variable temperature insert` (section VII A 1), and the insert is loaded into the cryostat (section VII E).

2. The cryostat's sample chamber is then flushed with helium (section VII C 1 and the heater chamber is evacuated (ref. VII B 2)

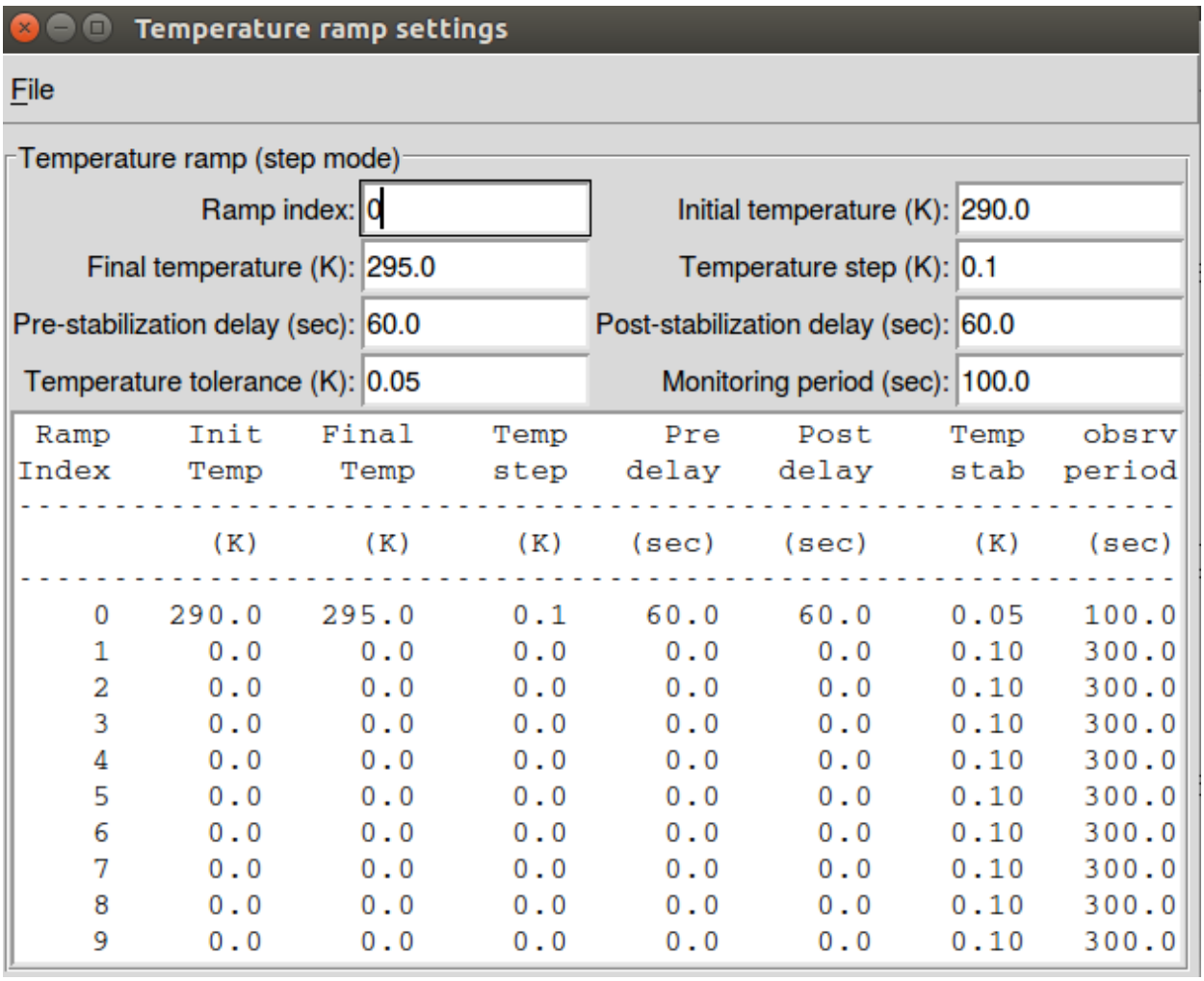
3. Next, liquid nitrogen is poured (ref. section VII D) to cool the sample temperature down to below 273.15 K.

4. To configure `Qrius` to measure resistance vs temperature (stepped ramp), navigate to `Measurement Mode` → `Electrical DC Conductivity Mode` from `Qrius`'s main menu. In the Electrical DC Conductivity window, select the `Run Mode` as `R-T (Stepped Ramp)`.

5. To configure resistance measurement settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `IV Source and Measurement Unit` →

`Resistance Measurement Settings`. Set the Current Limit to 10000 μA, Voltage Limit to 5000 mV, and Power Limit to 100 mW. Click on `File` → `Done` to save the settings.

6. To configure the temperature stepped ramp settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `Temperature Controller` → `Step Ramp Settings`. Configure the step ramp settings as shown in figure 16. Click on `File` → `Apply` to save the settings.



Ramp Index	Init Temp (K)	Final Temp (K)	Temp step (K)	Pre delay (sec)	Post delay (sec)	Temp stab (K)	obsrv period (sec)
0	290.0	295.0	0.1	60.0	60.0	0.05	100.0
1	0.0	0.0	0.0	0.0	0.0	0.10	300.0
2	0.0	0.0	0.0	0.0	0.0	0.10	300.0
3	0.0	0.0	0.0	0.0	0.0	0.10	300.0
4	0.0	0.0	0.0	0.0	0.0	0.10	300.0
5	0.0	0.0	0.0	0.0	0.0	0.10	300.0
6	0.0	0.0	0.0	0.0	0.0	0.10	300.0
7	0.0	0.0	0.0	0.0	0.0	0.10	300.0
8	0.0	0.0	0.0	0.0	0.0	0.10	300.0
9	0.0	0.0	0.0	0.0	0.0	0.10	300.0

FIG. 16. Temperature Stepped Ramp Settings [a]

[a] For a detailed explanation of the various parameters in the temperature stepped ramp settings, refer to section VIII C 1

7. Start the `R-T` run by clicking on the `Start` button on the `Electrical DC Conductivity` window. Wait for the acquisition to end. This may take a while depending on the number

of temperature setpoints at which data is being acquired.

8. Once the acquisition has ended, the resistance of the copper wire is plotted with respect to the `TCON`'s' sample temperature in figure 17.

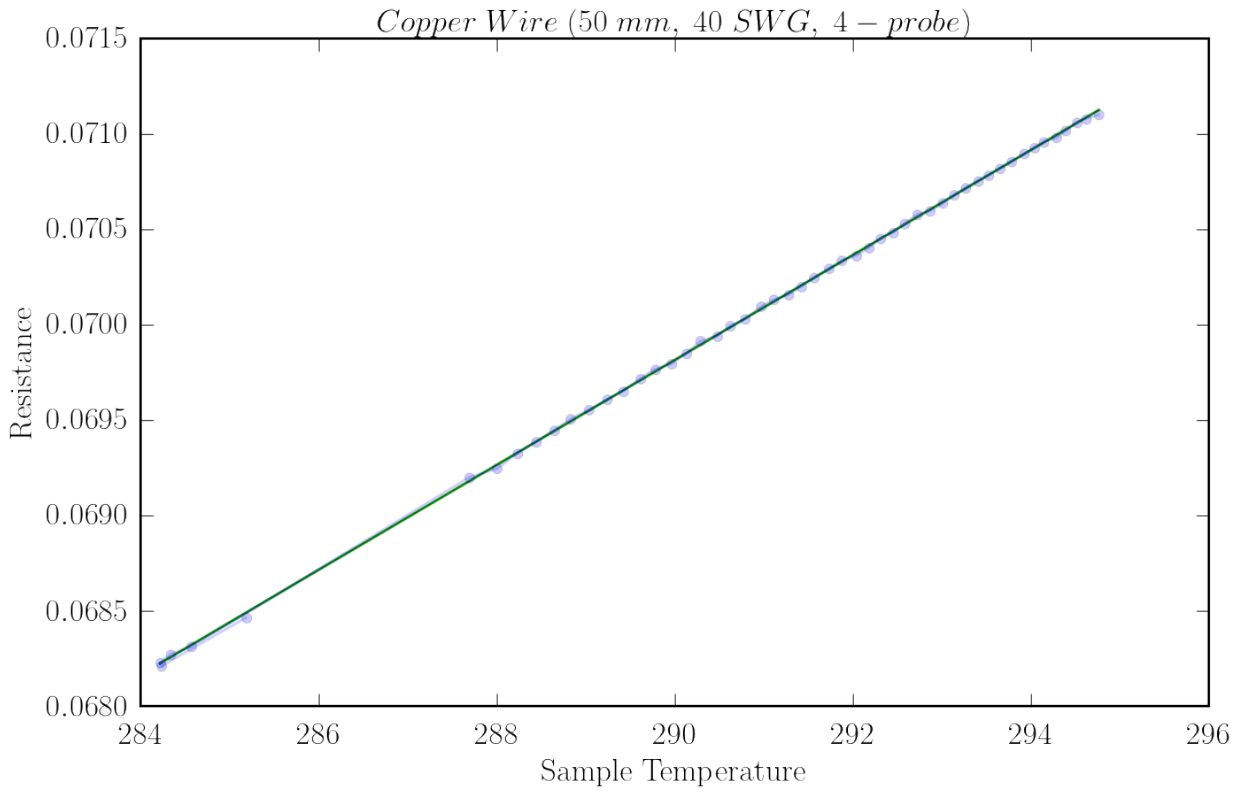


FIG. 17. Resistance vs Temperature

9. Because the geometry of the wire is known, we can calculate the resistivity from the resistance using the equation $\rho = RA/l$.

10. At 293.15 K, the slope of the ρ-T plot gives the temperature coefficient of resistivity for copper, denoted by α. Calculate its value and compare it to values presented in literature ($0.00393\ \Omega/K$).

11. Repeat the process with the 5 *cm* copper wire sample, to verify the reproducibility of the results with a different sample.

Conclusion : *Sample-1*

1. Calculated Resistance at 293.15 K, $R_{293.15K}$: _____ Ω

2. Measured Resistance at 293.15 K, $R_{293.15K}$: _____ Ω

3. Measured Resistivity at 293.15 K, $\rho_{293.15K}$: _____ $\Omega - m$

4. Absolute Error in $\rho_{293.15K}$: _____

5. Percentage Error in $\rho_{293.15K}$: _____

6. Temperature Coefficient of Resistance at 293.15 K, $\alpha_{293.15}$: _____ Ω/K

7. Absolute Error in $\alpha_{293.15}$: _____

8. Percentage Error in $\alpha_{293.15}$: _____

Sample-2

1. Measured Resistance at 293.15 K, $R_{293.15K}$: _____ Ω

2. Measured Resistivity at 293.15 K, $\rho_{293.15K}$: _____ $\Omega - m$

3. Absolute Error in $\rho_{293.15K}$: _____

4. Percentage Error in $\rho_{293.15K}$: _____

5. Temperature Coefficient of Resistance at 293.15 K, $\alpha_{293.15K}$: _____ Ω/K

6. Absolute Error in $\alpha_{293.15K}$: _____

7. Percentage Error in $\alpha_{293.15K}$: _____

V. INSULATORS

A. Measuring the Resistance of Insulating Materials

Objectives

1. To determine the resistance of an insulating sample

2. To measure the leakage currents flowing through an insulator at various voltages

Requirements

1. `XPLORE` High Resistance Measurement Unit

2. `XPLORE` Temperature Controller

3. `XPLORE` High Resistance Variable Temperature Insert

4. PZT pellet

5. Enamelled copper wire

6. Soldering iron and solder wire

Lead Zirconate Titinate or PZT is an inorganic ceramic compound with chemical composition $Pb[Zr_xTi_{1-x}]O_3$ where $0 > x > 1$. PZT is a dielectric, with a dielectric constant ranging from 300 to 3850 depending on its doping and orientation. PZT also posesses an interesting property known as piezoelectricity. It changes shape when a voltage is applied across it, and generates a voltage when mechanical pressure is applied to it. These properties makes PZT extremely useful in nano-positioning (in STMs and AFMs), precision position sensing, ultrasound transducers and ceramic capacitors.

In this experiment we measure the electrical resistance of the dielectric material, and characterize the leakage currents when voltages upto 100 V are applied across it. As discussed in section II A, we need a voltage source to probe samples with very high resistances. And because of their high resistance, a voltage applied across these samples results in a very tiny current (pA) flowing through them. In order to measure the resistance of such samples, we need to be able to measure these tiny values of currents.

Measurement of such miniscule currents requires specialized electronics, which minimize error in current measurement by reducing contributions from extraneous sources of current such as leakage currents through insulators within the instrument and the currents required to bias active components. The `SMU` has not been designed for ultra-low current measurements, and will thus saturate when trying to measure very high resistances. On the other hand, the `Hi-RES` module uses highly resistive, good quality insulators like teflon, *low bias current* amplifiers and *guarding*[3] to ensure low leakage currents, and measure ultra-low currents with precision. We use the `Hi-RES` and the `high resistivity variable temperature insert` in our endeavour to probe the current-voltage characteristics of a PZT sample.

1. The PZT pellet is mounted onto the `high resistivity variable temperature insert` (section VII A 2).

[3] A metal guard ring is placed around the lead which has to be guarded. The potential of the guard ring is (almost) equal to that of the lead it is guarding, which prevents any current from flowing between the two.

2. As the sample's dimensions exceed the maximum sample size that can possibly fit into the cryostat, we perform this experiment under ambient conditions. The insert (with the mounted sample) is kept on a non-conducting surface and left undisturbed for electrical measurements.

3. Go to `Modules Manager` → `High Resistance Measurement Unit`. To set the value of source voltage, set the `Value` field in `Settings` → `Source Parameters`. Click on `File` → `Apply`.

4. To measure the leakage current through the PZT sample when a voltage is applied across it, select `I-Time` from the `Run Mode` drop down menu located in the `High Resistance Measurement Unit` window.

5. Click on `Start` to begin acquisition. Wait for the current value to stabilize. The stable value of current being measured is the leakage current for our PZT Sample.

Conclusion : What can be said about the values of leakage current when 10V, 50V and 100V are applied across the sample?

Extra credit question: Ideal diodes allow the flow of current through them only in one direction. Practically, even when reverse biased, the diode conducts a small current through it, which is known as the reverse leakage current. A similar experiment may be performed on a diode in reverse bias to determine its reverse leakage current. The values stated in the diode's datasheet can then be compared against those obtained experimentally.

VI. SUPERCONDUCTORS

A. Characterizing the Superconducting Phase Transition

Objectives

1. To characterize the superconducting phase transition of a high temperature superconductor

Requirements

1. `XPLORE` Source and Measurement Unit

2. `XPLORE` Temperature Controller

3. `XPLORE` Low temperature cryostat with evacuation and flushing

4. `XPLORE` Variable Temperature Insert

5. `XPLORE` Sample Puck with a YBCO pellet attached

While studying the resistance of mercury at very low temperatures, Heike Kamerlingh Onnes discovered that below a temperature of 4.2 K, the resistance of mercury abruptly became zero. This phenomenon is known as superconductivity, which is very different from an ideal conductor that just exhibits zero resistivity.

As we have seen, the resistance of a metal decreases with temperature. Impurities and other defects limit this decrease in resistance in ordinary metals, meaning that even at 0 K, the metallic conductors have a small finite resistance (which is non-zero). On the other hand, the resistance drops abruptly to zero when the material is cooled below its critical temperature. The critical temperature is the temperature below which the phase transition from normal to superconducting occurs.

After Onnes' discovery of superconductivity in mercury, researchers observed the phenomenon in many other elements, with critical temperatures mainly in the range of below 1 K to 20 K. For conventional superconductors, the maximum critical temperature possible theoretically was 30 K. A breakthrough occurred when some cuprate-perovskite ceramic materials were discovered having critical temperatures above 90 K. The mechanism causing superconductivity in these ceramics is very different from conventional superconductors, which lead to the materials to be termed high-temperature superconductors. The properties of high temperature superconductors can be studied using liquid nitrogen which is cheaply available[4]. This facilitates many experiments and applications that are difficult to perform at lower temperatures.

In this experiment, we study the resistance vs temperature characteristics of a high-temperature super conductor - Yttrium Barium Copper Oxide $YBa_2Cu_3O_{7-\delta}$ - also known as YBCO. From the data obtained, we determine the critical temperature at which the transition from normal to superconducting state occurs, and the width of the transition region. We define the transition width, ΔT_c as the temperature range between 10% to 50% and 50% to 90% of the transition.

1. The sample puck with the YBCO pellet[5] is mounted onto the `variable temperature insert` (section VII A 1), and the insert is loaded into the cryostat (section VII E).

[4] In New Delhi, liquid nitrogen costs range from Rs.50-60 per litre, as compared to about Rs.1000 per litre for liquid helium

[5] The sample puck is first covered with a layer of Kapton tape onto which the YBCO pellet is stuck using varnish. The layer of tape is required to electrically insulate the YBCO sample from the copper block underneath. Electrical contacts to the YBCO are made in linear 4-probe configuration with thin copper wires stuck to the YBCO pellet using silver ink.

2. The cryostat's sample and heater chambers are then flushed with helium (section VII C 1 and VII C 2

3. Next, liquid nitrogen is poured (ref. section VII D) to cool the sample temperature down to about 80 K.

4. To configure `Qrius` to measure resistance vs temperature (stepped ramp), navigate to `Measurement Mode` → `Electrical DC Conductivity Mode` from `Qrius`'s main menu. In the Electrical DC Conductivity window, select the `Run Mode` as `R-T (Stepped Ramp)`.

5. To configure resistance measurement settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `IV Source and Measurement Unit` → `Resistance Measurement Settings`. Set the Current Limit to 10000 μA, Voltage Limit to 5000 mV, and Power Limit to 100 mW. Click on `File` → `Done` to save the settings.

6. To configure the temperature stepped ramp settings, go to the `Settings` drop-down menu in the `Electrical DC Conductivity` window, and select `Temperature Controller` → `Step Ramp Settings`. Configure the step ramp settings as shown in figure 18. Click on `File` → `Apply` to save the settings.

7. Start the `R-T` run by clicking on the `Start` button on the `Electrical DC Conductivity` window. Wait for the acquisition to end. This may take a while depending on the number of temperature setpoints at which data is being acquired.

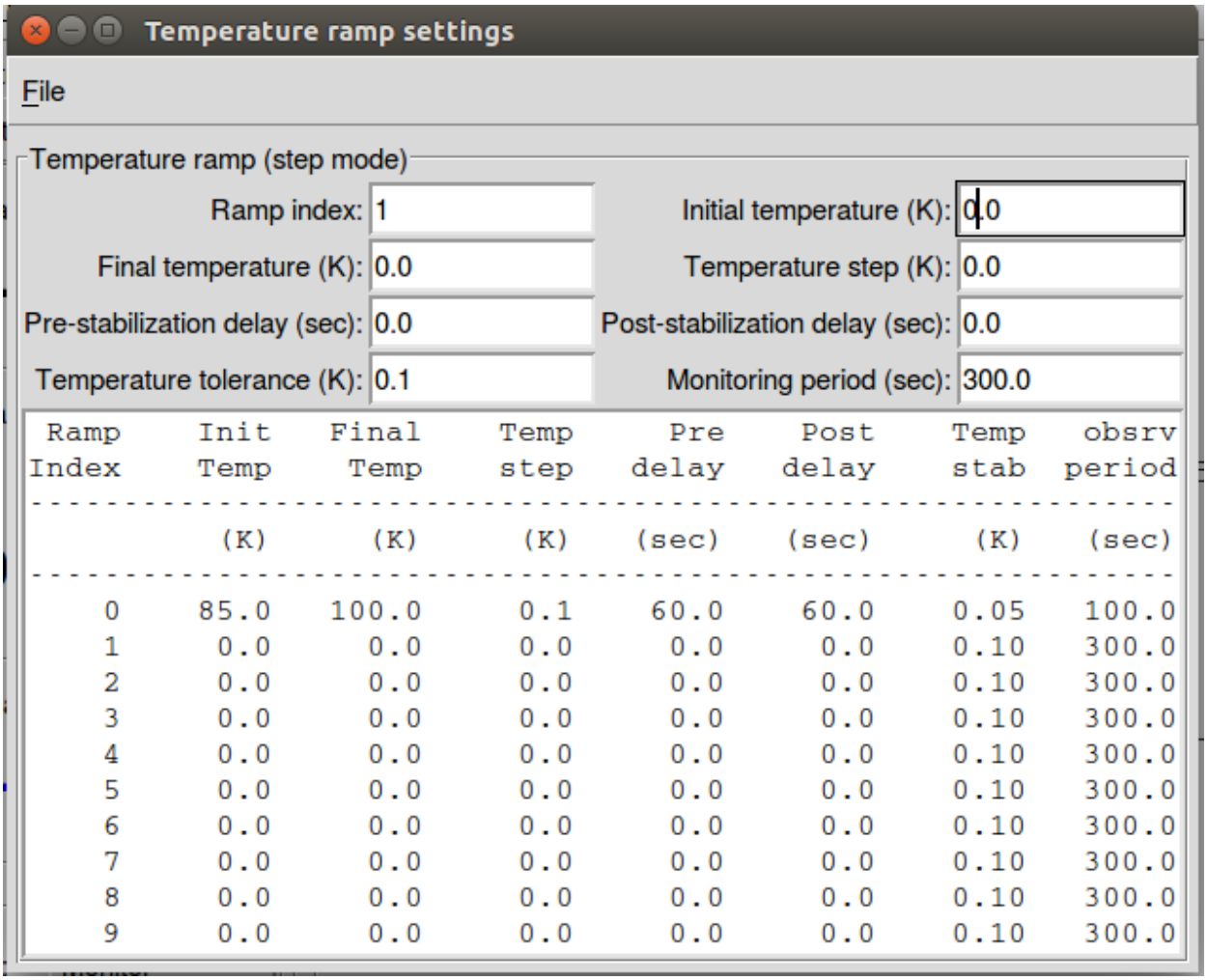
8. Once the acquisition has ended, the resistance of the YBCO pellet is plotted with respect to the `TCON`'s sample temperature in figure 19.

Conclusion

1. Critical Temperature, T_c : _____ K

2. Transition Width, ΔT_c : _____ K

3. Full Resistance, R_f : _____ Ω



Ramp Index	Init Temp (K)	Final Temp (K)	Temp step (K)	Pre delay (sec)	Post delay (sec)	Temp stab (K)	obsrv period (sec)
0	85.0	100.0	0.1	60.0	60.0	0.05	100.0
1	0.0	0.0	0.0	0.0	0.0	0.10	300.0
2	0.0	0.0	0.0	0.0	0.0	0.10	300.0
3	0.0	0.0	0.0	0.0	0.0	0.10	300.0
4	0.0	0.0	0.0	0.0	0.0	0.10	300.0
5	0.0	0.0	0.0	0.0	0.0	0.10	300.0
6	0.0	0.0	0.0	0.0	0.0	0.10	300.0
7	0.0	0.0	0.0	0.0	0.0	0.10	300.0
8	0.0	0.0	0.0	0.0	0.0	0.10	300.0
9	0.0	0.0	0.0	0.0	0.0	0.10	300.0

FIG. 18. Temperature Stepped Ramp Settings [a]

[a] For a detailed explanation of the various parameters in the temperature stepped ramp settings, refer to section VIII C 1

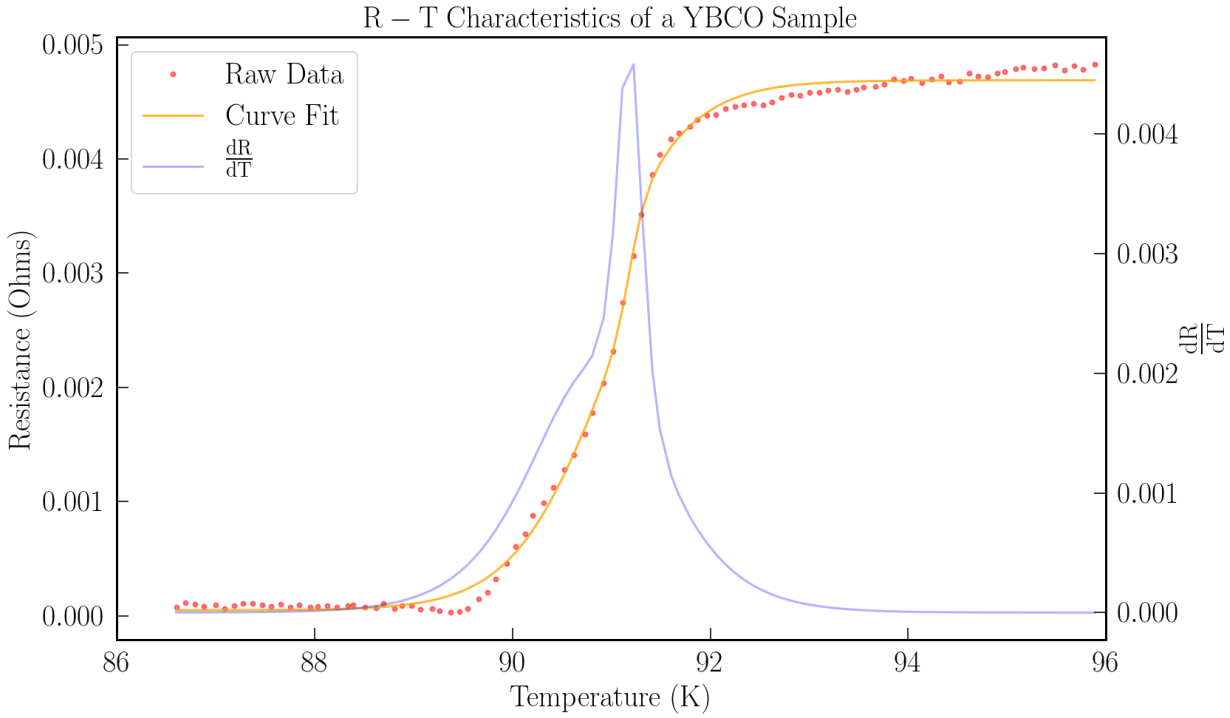


FIG. 19. YBCO Resistance vs Temperature

VII. OPERATIONAL PROCEDURES

A. `mount_sample(insert)` : Mount a sample on the chosen `insert`

Measurement of current-voltage characteristics of a device requires stable Ohmic contacts which go all the way from the sample to the electronics measurement boxes. For this purpose, `XPLORE` offers a variety of inserts, which . The choice of sample and measurements together dictate the insert that will be used to perform those measurements.

1. `Variable Temperature Insert`

The `variable temperature insert` provided with `XPLORE` 1.2 is well suited for samples with resistances in the range of 1 $m\Omega$ to 100 $M\Omega$. The insert comes along with a detachable `sample puck`, on which the sample is mounted. It is designed to facilitate quick and easy sample changes, and is capable of withstanding temperatures in the range of 77-450K.

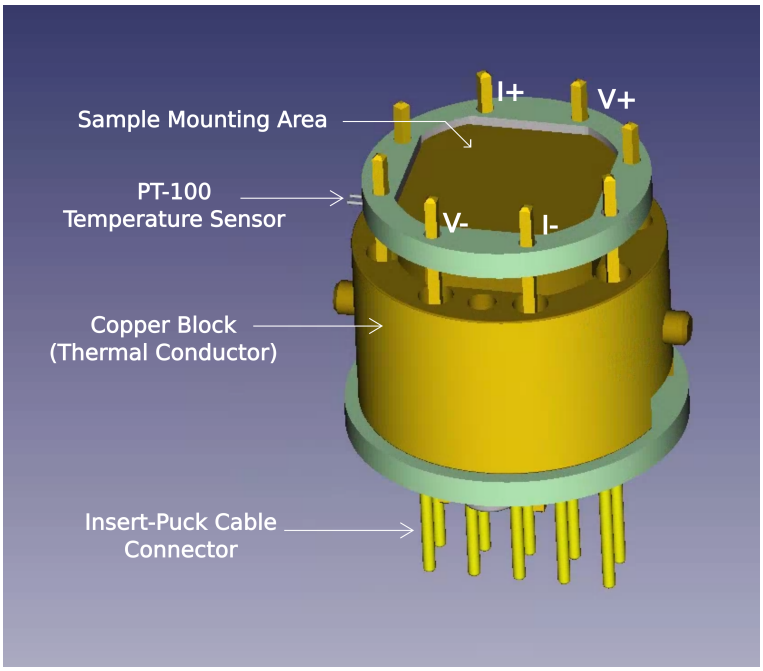


FIG. 20. Sample Puck for Variable Temperature (77-450 K) Insert

The sample is soldered on the `sample puck`, which is a detachable platform that provides electrical connections from the insert to the sample (as shown in fig. 21).

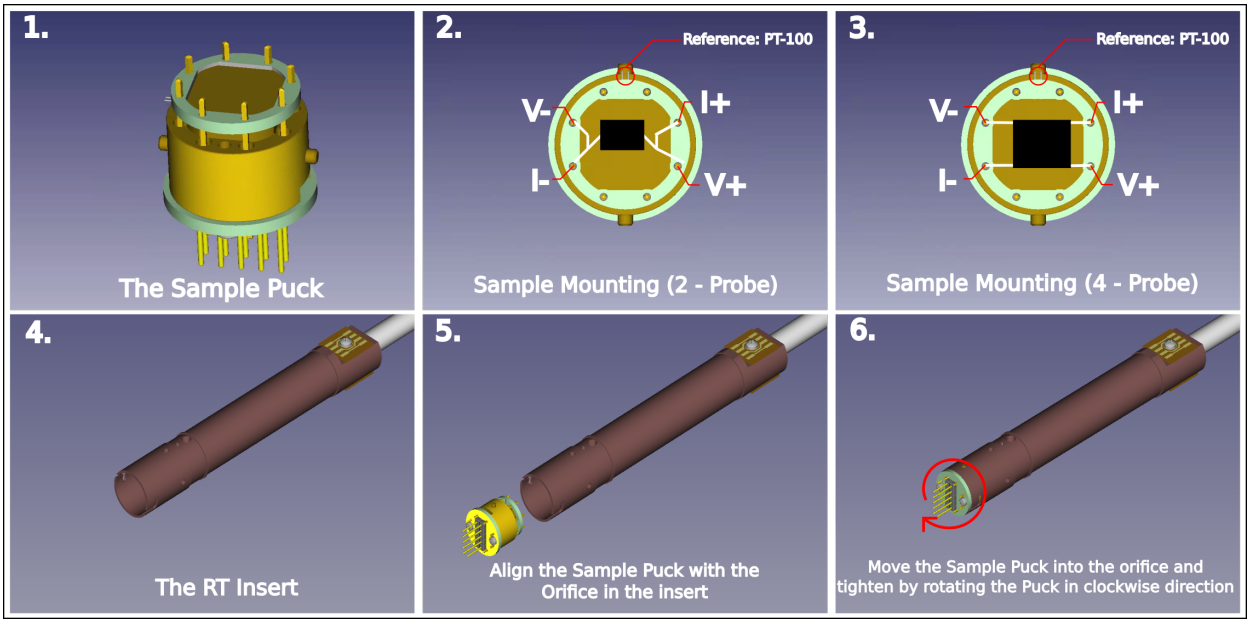


FIG. 21. Mounting the Sample on the R-T Insert

Note :

While mounting an electrically conductive sample, care must be taken to insulate the surface of the sample from the surface of the puck using a layer of Kapton tape.

Once the sample is mounted on the `sample puck`, the puck is screwed on to the `insert`. Electrical connections from the puck to the insert are made using the `FRC` connectors provided on both parts.

Note :

Remember to align the `FRC` connector's extruded block with corresponding marking on the back of the `sample puck`'s PCB.

2. `High Resistivity Variable Temperature Insert`

This insert has been designed for low leakage current measurements, suitable for samples with resistances in the range of 1 $M\Omega$ to 100 $T\Omega$. It is capable of withstanding temperatures in the range 77-450K.

Samples are mounted on the insert directly by soldering the two terminals of the sample with those of the insert.

Note :

To maintain the integrity of low leakage currents in precision measurements, the `high resistivity variable temperature insert` should be protected from dust and moisture. To remove moisture deposited on the insert, it is recommended to heat up the insert in vacuum inside the cryostat prior to performing the measurements.

B. `evacuate(chamber)` : Evacuate the chosen `chamber`

Note :

Always evacuate the sample and heater chamber when the system is not in use! This reduces the accumulation of moisture and dust inside the cryostat, and helps in longer storage of cryogen (liquid nitrogen) in the cryocan by providing a thermally insulating layer.

1. `Sample_Chamber`

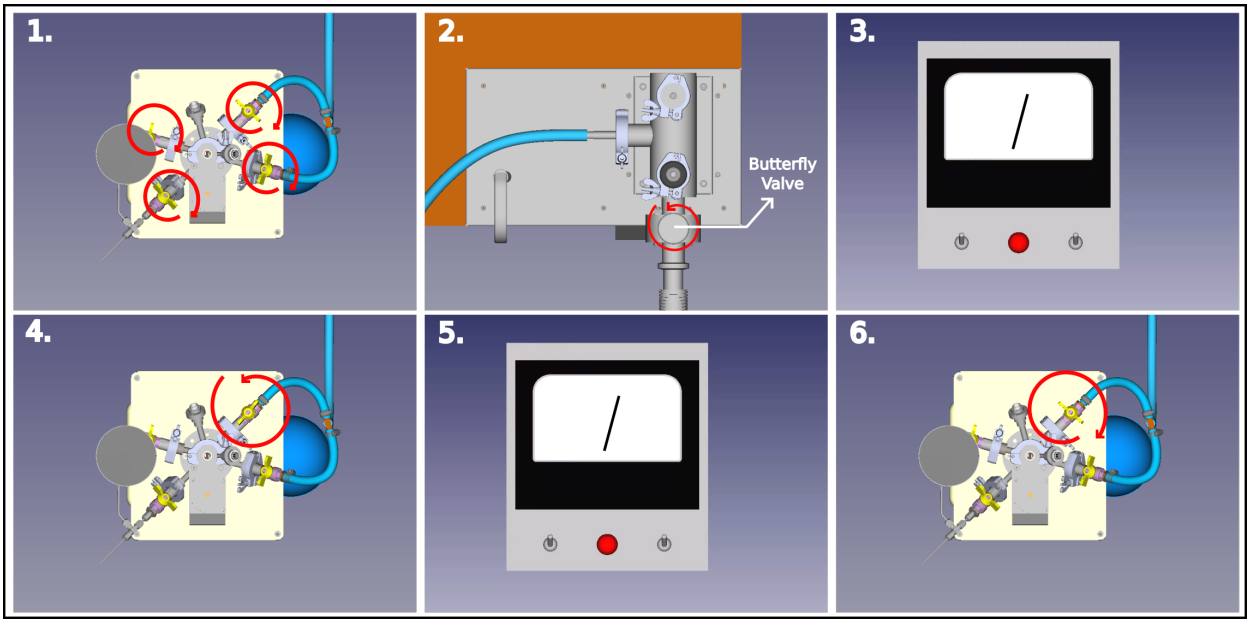


FIG. 22. Evacuating the Sample Chamber

Fig. 22 illustrates the sequence of steps required to evacuate the sample chamber. Further explanation of each step is provided below :

1. Prior to evacuation, ensure that all the valves are in the closed position. As a rule of thumb, valves are in a closed state when they are perpendicular to the inlet pipe; and open when they are parallel to the piping.

2. To evacuate the vacuum manifold, open the pump's butterfly valve by rotating it in the anticlockwise direction. All valves in XPLORE open when rotated in the anticlockwise direction and close when they are rotated in a clockwise manner.

3. Wait for stable vacuum to be achieved. Let the Pirani gauge needle stabilize.

4. To begin evacuation of the sample chamber, open the sample chamber evacuation valve by rotating in anticlockwise direction as shown in step (4) of fig. 22.

5. Wait for the Pirani gauge needle to stabilize.

6. Once stable vacuum has been achieved in the sample chamber, close the sample chamber evacuation valve by rotating in clockwise direction.

2. Heater_Chamber

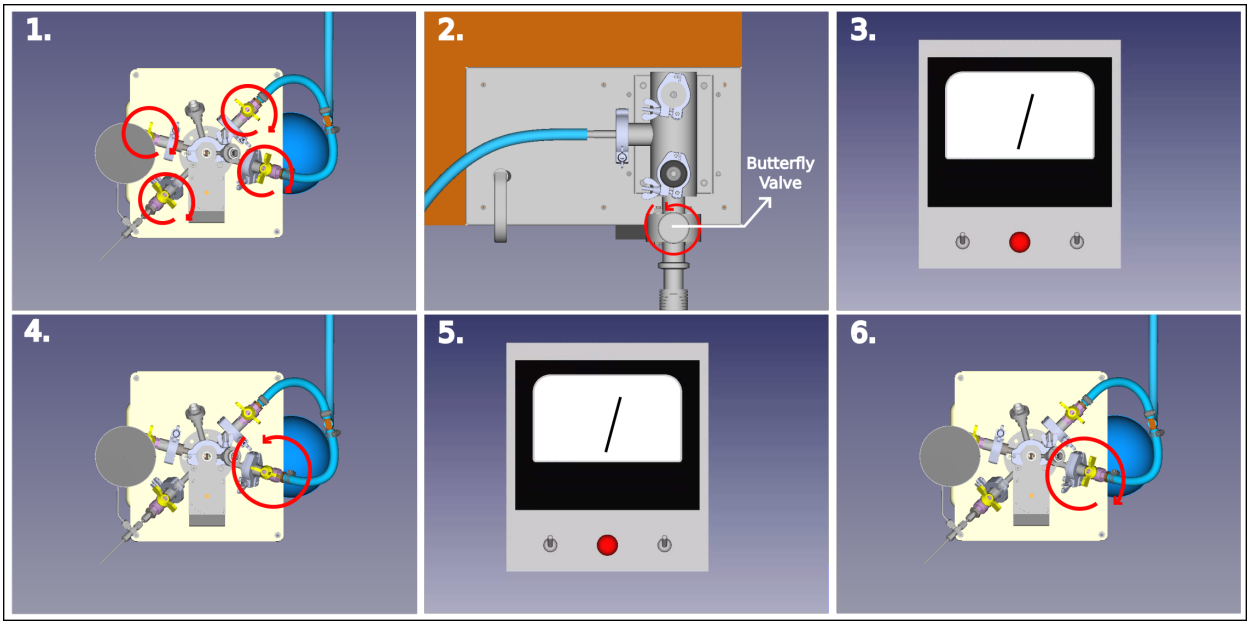


FIG. 23. Evacuating the Heater Chamber

The procedure to evacuate the Heater Chamber is illustrated in fig. 23. Further explanation of each step is provided below :

1. Prior to evacuation, ensure that all the valves are closed (perpendicular to the inlet pipe, rotated maximally in clockwise direction).

2. To evacuate the vacuum manifold, open the pump's butterfly valve by rotating in anticlockwise direction.

3. Wait until stable vacuum has been achieved in the manifold. Let the Pirani gauge needle stabilize.

4. To evacuate the heater chamber, open the heater chamber evacuation valve (shown in step (4) of fig. 23) by rotating in the clockwise direction.

5. Wait for the Pirani gauge needle to stabilize.

6. Close the heater chamber evacuation valve by rotating in clockwise direction.

C. `flush_helium(chamber)` : Flush the chosen `chamber` with Helium

In order to cool the sample down, the heater and sample chambers must be flushed with an exchange gas such as helium. Due to its high thermal conductivity, helium helps the transfer of heat from the sample into the liquid nitrogen, cooling the sample down in the process.

While heating up the sample, we need to evacuate the heater chamber to cut off the flow of heat from the sample to liquid nitrogen. Vacuum creates a thermally insulating layer, which makes it easier for the heater to heat up the sample. The sample chamber can be kept flushed with helium even while heating, in order to minimize temperature gradients across the sample inside it. This helps ensure that the sample and the temperature sensor (which are separated by about a millimeter) experience the same temperatures.

Note :

Prior to flushing any chamber, ensure that an insert/cover is in place over the cryostat's orifice and is securely fastened by a clamp.

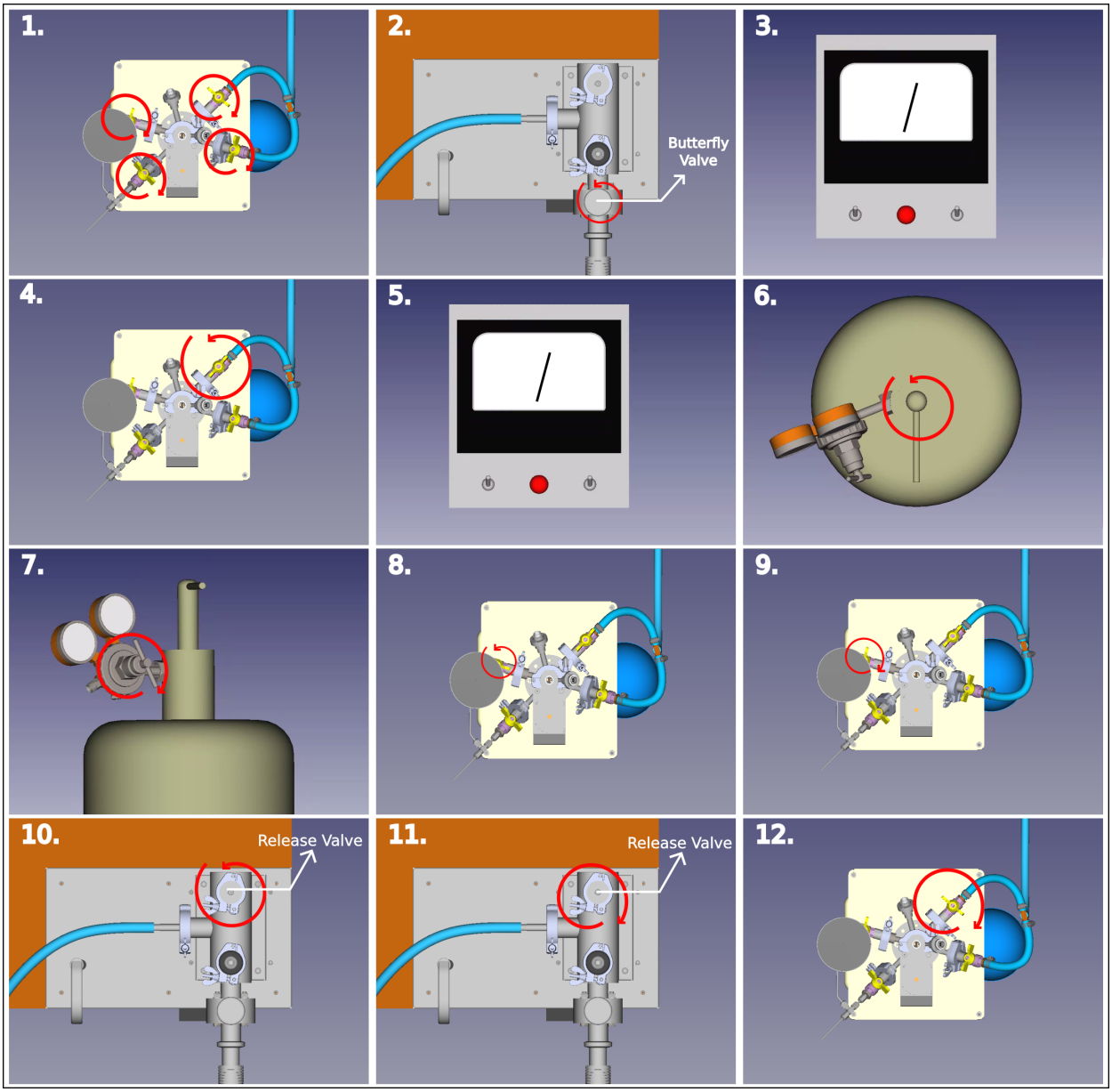


FIG. 24. Flushing the Sample Chamber with Helium gas

1. `Sample_Chamber`

Fig. 24 illustrates the sequence of steps required to flush the sample chamber with an exchange gas (helium in this case). Each of the steps are explained below :

1. Before initiating the flushing process, ensure that all the valves are closed (maximally rotated in the clockwise direction) as shown in step (1) of fig. 24.

2. Evacuation of the sample chamber is necessary before flushing helium into it. To start

evacuating the vacuum manifold, open the pump's butterfly valve by rotating in anticlockwise direction.

3. Wait for stable vacuum to be achieved in the vacuum manifold. Let the Pirani gauge needle stabilize.

4. To start the evacuation of the sample chamber (the chamber into which helium is to be flushed), rotate the sample chamber's evacuation valve in the anticlockwise direction.

5. Wait for the Pirani gauge needle to stabilize. While the Pirani gauge needle stabilizes, we can set up the helium source.

6. Turn on the helium cylinder's main valve by rotating in anticlockwise direction.

7. To set the output pressure from the helium cylinder, slowly turn the pressure regulator in the clockwise direction. Set the pressure to be about 10-15 psi.

8. After stable vacuum has been achieved in the sample chamber and the helium source has been set up, we proceed to flush the sample chamber with helium. To do this, first cut-off the vacuum pump by rotating the pump's butterfly valve in clockwise direction. Then open the sample chamber's flush valve by turning it in the anticlockwise direction.

9. Wait for 2-3 seconds for the helium to flow into the sample chamber. Then close the sample chamber's flush valve by turning it in the clockwise direction.

10. Now excess pressure built up due to helium inside the sample chamber must be released in order to prevent damage to the o-rings. To release the excess pressure, open the release valve (on the vacuum manifold) by turning it in the anticlockwise direction until a sharp hissing sound (caused by the escaping gas) is heard.

11. Close the release valve by rotating it maximally in the clockwise direction.

12. Finally, to confine the helium gas to the sample chamber, the evacuation valve is closed by rotating it in the clockwise direction as shown in step (12) of fig. 24. The pump's butterfly valve may now be opened by rotating in anticlockwise direction (after ensuring that all valves shown in step (1) of fig. 24 are closed). To prevent leakage, turn off the helium source by turning the cylinder's main valve in clockwise direction.

2. `Heater_Chamber`

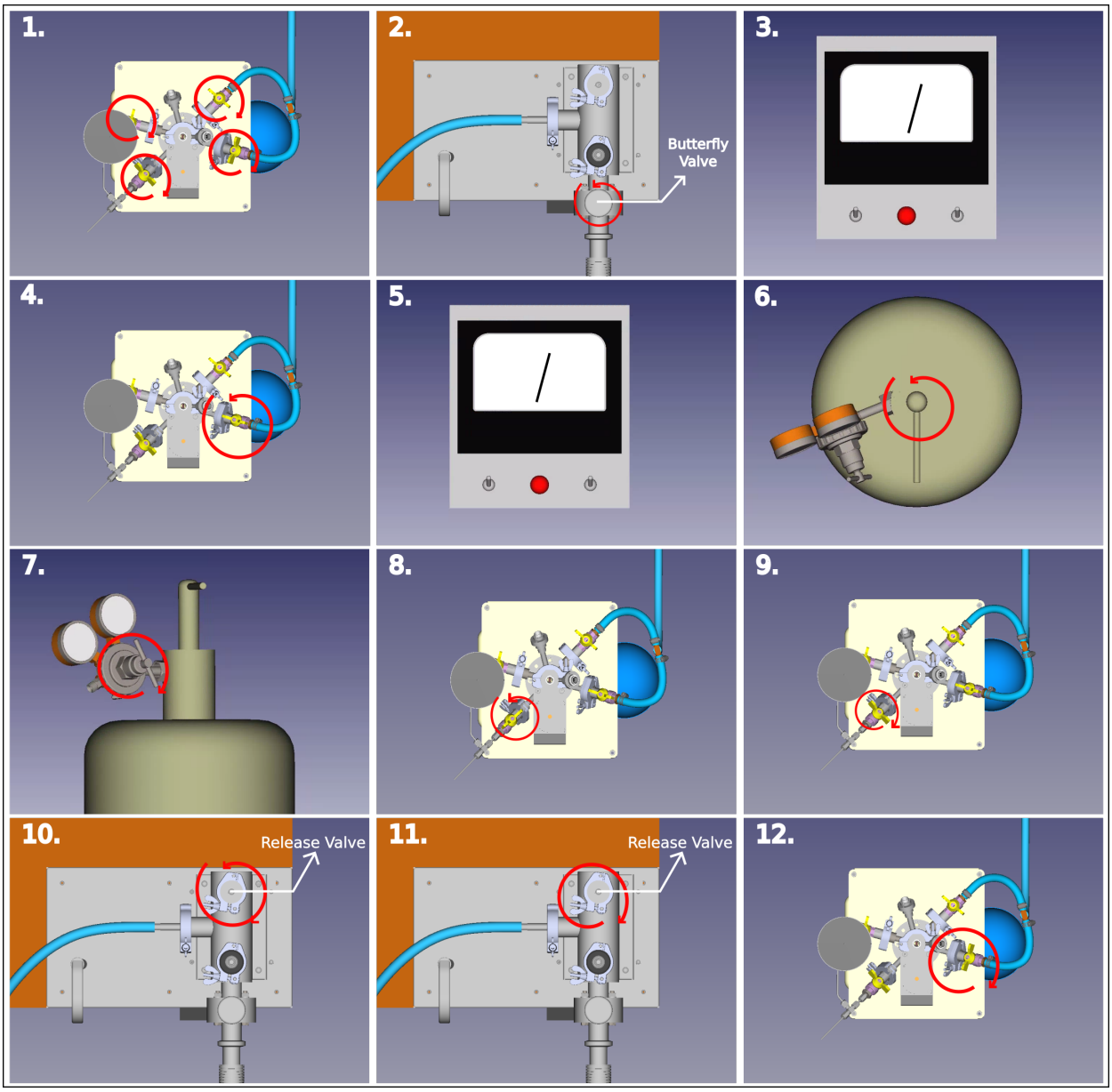


FIG. 25. Flushing the Heater Chamber with Helium gas

Fig. 25 illustrates the procedure to flush the heater chamber with Helium gas. Each step is detailed below :

1. Before initiating the flushing process, ensure that all the valves are closed (maximally rotated in the clockwise direction) as shown in step (1) of fig. 25.

2. Evacuation of the heater chamber is necessary before flushing helium into it. To start evacuating the vacuum manifold, open the pump's butterfly valve by rotating in anticlockwise direction.

3. Wait for stable vacuum to be achieved in the vacuum manifold. Let the Pirani gauge needle stabilize.

4. To start the evacuation of the heater chamber (the chamber into which helium is to be flushed), rotate the heater chamber's evacuation valve in the anticlockwise direction.

5. Wait for the Pirani gauge needle to stabilize. While the Pirani gauge needle stabilizes, we can set up the helium source.

6. Turn on the helium cylinder's main valve by rotating in anticlockwise direction.

7. To set the output pressure from the helium cylinder, slowly turn the pressure regulator in the clockwise direction. Set the pressure to be about 10-15 psi.

8. After stable vacuum has been achieved in the heater chamber and the helium source has been set up, we proceed to flush the heater chamber with helium. To do this, first cut-off the vacuum pump by rotating the pump's butterfly valve in clockwise direction. Then open the heater chamber's flush valve by turning it in the anticlockwise direction.

9. Wait for 2-3 seconds for the helium to flow into the heater chamber. Then close the heater chamber's flush valve by turning it in the clockwise direction.

10. Now excess pressure built up due to helium inside the heater chamber must be released in order to prevent damage to the o-rings. To release the excess pressure, open the release valve (on the vacuum manifold) by turning it in the anticlockwise direction until a sharp hissing sound (caused by the escaping gas) is heard.

11. Close the release valve by rotating it maximally in the clockwise direction.

12. Finally, to confine the helium gas to the heater chamber, the evacuation valve is closed by rotating it in the clockwise direction as shown in step (12) of fig. 25. The pump's butterfly valve may now be opened by rotating in anticlockwise direction (after ensuring that all valves shown in step (1) of fig. 25 are closed).To prevent leakage, turn off the helium source by turning the cylinder's main valve in clockwise direction.

D. `pour_liquid_nitrogen()` : Cool the Cryostat with Liquid Nitrogen

To cool the sample below room temperature, liquid nitrogen is required to be be poured into the dewar containing the `XPLORE` cryostat. The cryostat is generally cooled down after the sample has been loaded into the cryostat, and the necessary flushing has been performed.

1. To monitor the cooling rate, we set up the temperature controller in `Monitor` mode. In `Modules Manager`, click on `Temperature Controller`. Inside the `Temperature Controller` window, select `Monitor` from the `Run Mode` drop down menu. Click on `Start`.

2. Liquid nitrogen is slowly poured into the funnel. The temperature on the `Monitor` plot will start decreasing.

3. Stop pouring liquid nitrogen once the minimum required temperature has been reached (or if the cryocan starts over flowing).

4. To end the `Monitor` run, click on the `Stop` button in the `Temperature Controller` window.

E. `load_insert()` : Load an Insert into the Cryostat

In order to make high precision measurements along with precise control over its temperature, a sample is mounted on a variable temperature insert, which is then loaded into the cryostat as described in fig. 26.

Prior to loading the insert into the cryostat, helium gas is flushed into the sample chamber (refer to the flushing procedure in section VII C 1).

Next, the clamp holding the cryostat's cover in place is unfastened and the cover is removed. The insert is now gently inserted vertically downward completely, and the clamp is tightened back into place.

Note :

Flushing helium while loading/unloading an insert prevents the condensation of atmospheric water vapour inside the cryostat. Moisture causes damage to the cryostat!

F. `unload_sample()` : Unload an Insert from the Cryostat

To unload an insert from a cryostat (fig. 27), the sample chamber is flushed with helium as mentioned section VII C 1.

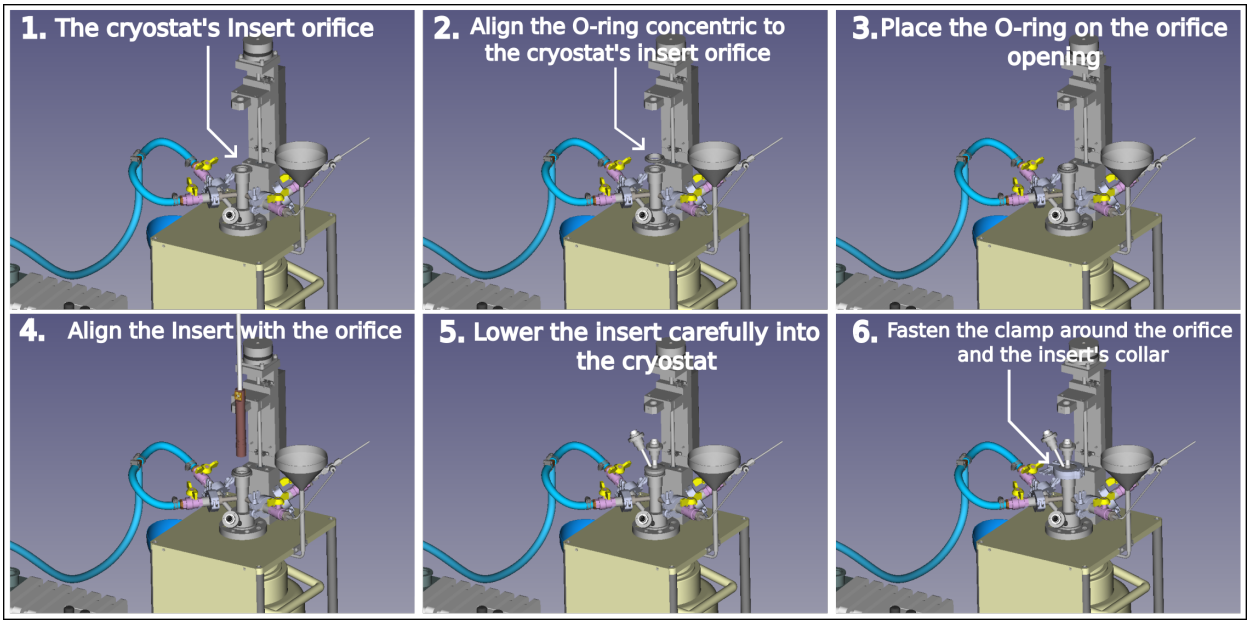


FIG. 26. Loading an Insert into the Cryostat

Next, the clamp holding the insert in place is removed and the insert is carefully removed from the cryostat by moving in a vertically upward direction.

The cryostat's cover is replaced over the opening, and the clamp is fastened over it.

Note :

To keep the cryostat protected from dust and moisture, always cover the cryostat's opening using an insert or its cover, fastening it firmly in place with a clamp.

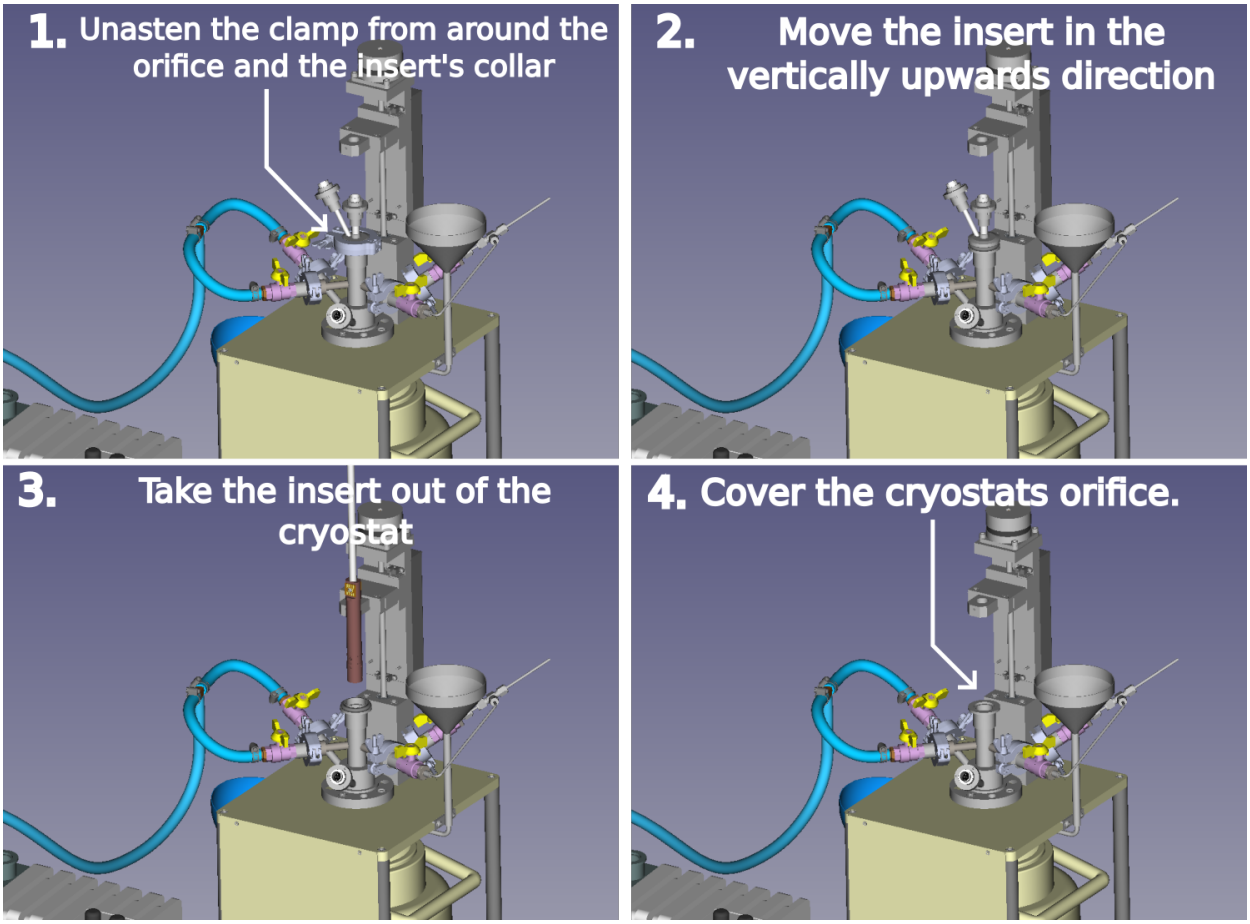


FIG. 27. Unloading an Insert from the Cryostat

VIII. QRIUS : CONTROL AND DATA ACQUISITION FOR XPLORE

`Qrius` is a clean and intuitive graphical user interface to the `XPLORE` Physical Quantities Measurement System that provides an easy way to control and monitor your experiment. We understand that the researcher's valuable time needs to be spent in designing the experiment and analysis of the data rather than in manually jotting down numbers while performing an experiment. `Qrius` makes it possible to completely automate control of the various experimental parameters as well as data acquisition. Settings can be changed with the click of a button and live plots are rendered as the experiment runs.

In this section we discuss the layout of the graphical user interface and make sense of some commonly used settings provided in `Qrius`.

A. Introduction : Design and Layout

`Qrius`'s main window is shown in fig. 28. This window is divided into 4 major sections based on functionality. The toolbar at the top provides control over connectivity with instruments (`Init`), plotting facilities (`Open`), access to the instrument manual (`Help`), and controlled shutdown of the software `Exit`. `Sample Settings` lets the user organize data by samples. `Modules Manager` allows control over individual modules of the `XPLORE` system, while `Measurement Modes` controls and acquires data from multiple modules simultaneously, making it easy to run complex experiments and extract useful information out of them.

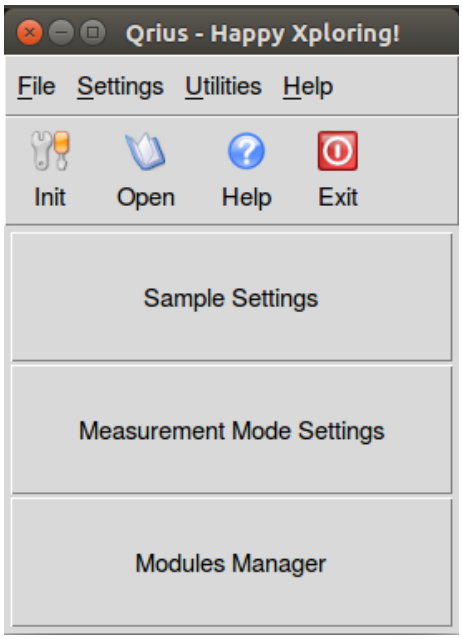


FIG. 28. `Qrius` main window

B. Saving and Plotting Data

Apart from live plots, data is stored in `.csv` and `.xpl` formats, complete with time stamps for each data point. The default nomenclature of the file is in the format `yyyymmdd_time_runmode.ext`.

1. Data Folder

Data files are saved in the `Data Folder`, which can be changed from `Settings` → `Global Settings` in `Qrius`'s main window.

2. Sample Settings

To organize the data based on samples, the `Sample Settings` may be used which allows for separate directories for each sample ID inside the `Data Folder`. Detailed information about a sample like batch number, ID number, preparation processes, dimensions, etc can be stored in the `Sample Description` field.

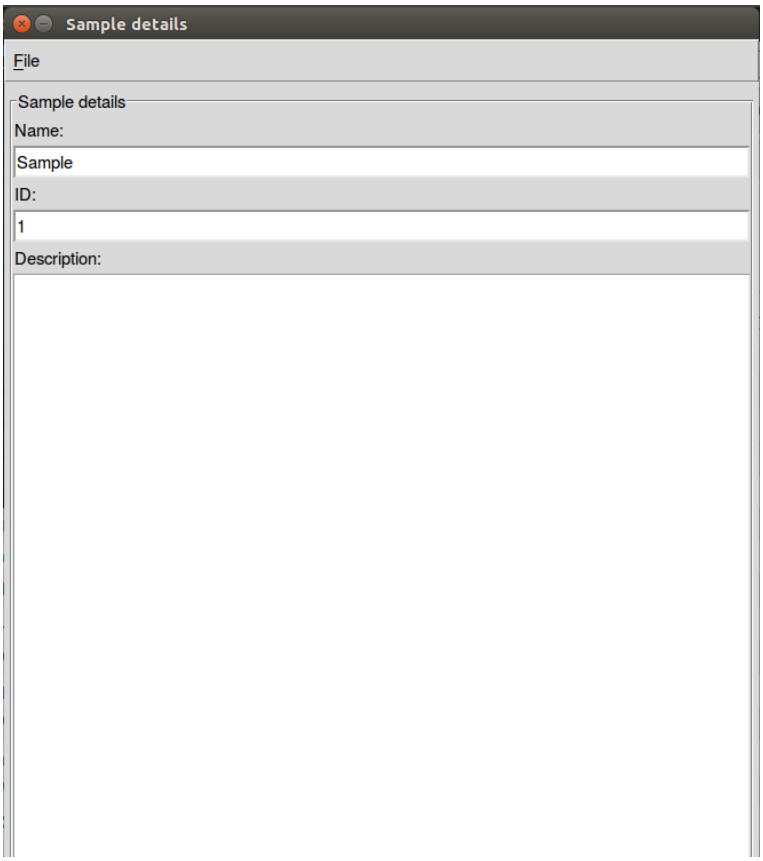


FIG. 29. `Qrius` Sample Settings

3. Generating Plots

`.xpl` files stored as described above can be plotted by the `Qrius`'s plotting utility by clicking on `Open` in the main window. The required file is then loaded by clicking on `File` → `Import`. The data to be represented by each axis is selected. Clicking on the `Add` button generates a plot of the chosen parameters in data file.

C. Modules Manager

This section lets the user control and acquire data from individual modules in the `XPLORE` system (ref. 30).

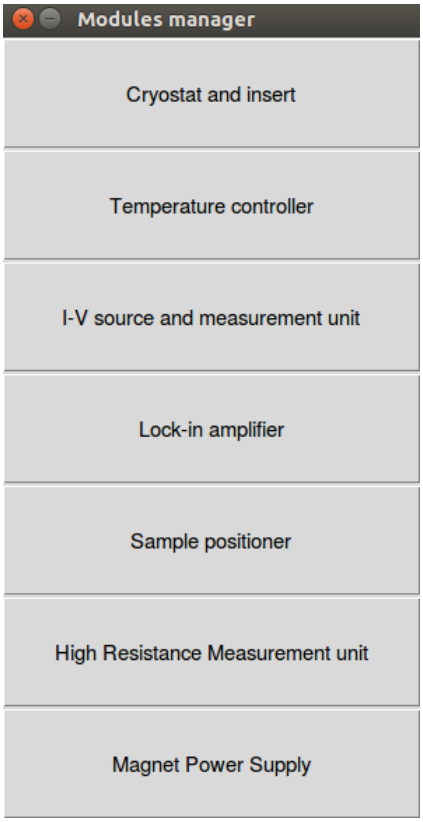


FIG. 30. `Qrius` Modules Manager

1. *Temperature Controller*

The temperature controller module (`TCON`) in `XPLORE` is a microprocessor based, computer controlled module which allows the sample to be subjected to a variable temperature range (80-450 K), with a temperature stability better than 0.1 K and a temperature measurement resolution of 0.01 K. It has two temperature measurement channels and a power output channel to drive a 35 Watt heater. The system can be calibrated to work with standard platinum RTDs (PT100) and thermocouples.

The TCON can be configured to run in one of the following modes :

1. Monitor : Temperature readings from both the sample and heater temperature sensors are

plotted with respect to time in this run mode when the `Start` button is pressed. Temperature is not controlled, and the power supplied to the heater remains zero during this mode.

2. Isothermal : A desired temperature is set by clicking on `Settings` → `Isothermal Settings`, and entering the value of temperature setpoint. When the `Start` button is pressed on the `Temperature Controller` window, the heater is powered using a PID control loop to maintain constant temperature (setpoint). Both sample and heater temperatures are plotted with time.

3. Linear Ramp : The temperature is linearly ramped up/down starting from the current temperature and ending at the set final temperature. Linear ramp settings are configured by clicking on the `Settings` → `Linear Ramp Settings`, and setting the `Ramp Rate` and `Final Temperature`.

   Parameters

   (a) Final Temperature : The temperature is ramped up/down starting from the current temperature, ending at the set final temperature.

   (b) Ramp Rate : Rate at which the temperature is ramped up/down.

4. Stepped Ramp : The temperature is incremented in steps, and stabilized after each step. The step ramp settings can be configured from `Settings` → Step Ramp Settings in the `Temperature Controller` window.

   Parameters

   (a) Ramp Index : Multiple temperature stepped ramps can be programmed to run one after the other. The ramp index indicates the serial number of a particular ramp.

   (b) Initial Temperature : Temperature (in Kelvin) from which the ramp begins.

   (c) Final Temperature : Temperature (in Kelvin) at which the ramp ends.

   (d) Temperature Step : The temperature is incremented in multiples of temperature step.

   (e) Pre-stabilization Delay : Time delay (in seconds) before stabilization begins.

   (f) Post-stabilization Delay : Time delay (in seconds) after stabilization ends.

   (g) Temperature Tolerance : The temperature is deemed stable if it stabilizes to within the temperature tolerance over the monitoring period.

(h) Monitoring Period : The time period (in seconds) over which temperature fluctuations are monitored to ensure temperature stability.

The parameters in the step ramp settings are illustrated in figure 31

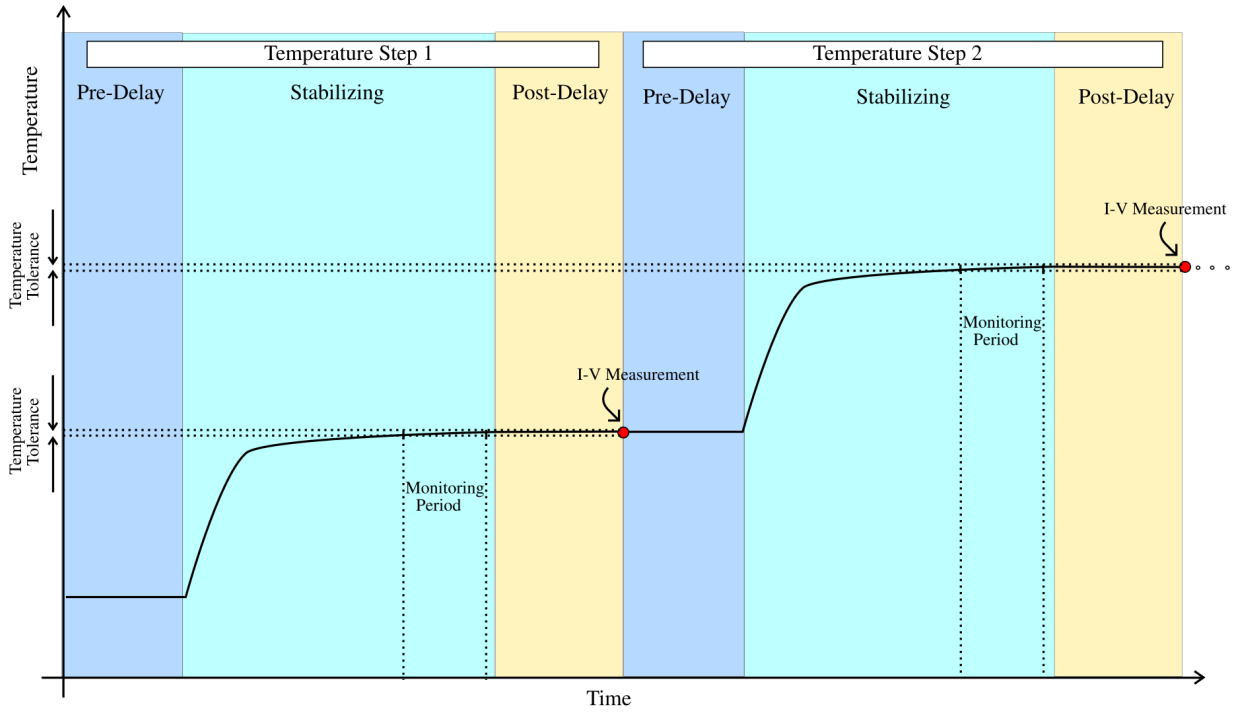


FIG. 31. `Qrius` Temperature Step Ramp Settings

2. Source and Measurement Unit

The `XPLORE` Source-Meter Unit (`SMU`) can supply a constant current or voltage, and measure current and voltages with high precision as a function of time. The `Intelligent Switching` algorithm implemented in `Qrius` allows the `SMU` to dynamically switch between current and voltage source modes making it possible to study devices with highly non-linear I-V characterisics.

Apart from measurement of currents and voltages vs time, the `SMU` can run in the following modes :

1. R-Time : Resistance of the sample is measured with respect to time. The resistance measurement settings can be configured from `Settings` → `Resistance Measurement Settings` in the `Source and Measurement Unit` window.

   Parameters

(a) Current Limit : The maximum current (in μA) allowed to pass through the sample, above which acquisition is ended.

(b) Voltage Limit : The maximum voltage (in mV) across the sample, above which acquisition is ended.

(c) Power Limit : The maximum power (in mW) allowed to dissipate across the sample, above which the acquisition is ended.

(d) Bipolar : When set to `Yes`, the resistance is measured once, after which the source polarity is switched and resistance is measured again. The two resistance values are averaged to cancel out thermoelectric voltages.

2. I-V : Complete I-V curve of the sample is measured and plotted. Depending on the resistance of the sample, the source is automatically switched between current/voltage. The I-V measurement settings can be configured from `Settings` → `I-V Measurement Settings` in the `Source and Measurement Unit` window.

   Parameters

   (a) Current Limit : The maximum current (in μA) allowed to pass through the sample, above which acquisition is ended.

   (b) Voltage Limit : The maximum voltage (in mV) across the sample, above which acquisition is ended.

   (c) Power Limit : The maximum power (in mW) allowed to dissipate across the sample, above which the acquisition is ended.

   (d) Current Step Size : The increment in drive current (in μA) when current source mode is active.

   (e) Voltage Step Size : The increment in source voltage (in mV) when voltage source mode is active.

   (f) Bipolar : When set to `Yes`, the I-V characteristics are measured both in the 1st and 2nd quadrants of the I-V space. When set to `No`, the measurement takes place only in the 1st quadrant.

Acquisition Settings can be used to control the time delay after excitation (and before acquisition), and the internal averaging of data points. These settings can be configured inside the

`I-V Source and Measurement Unit` window by clicking on `Settings` → `Acquisition Settings`. The parameters are explained below :

1. Delay : All measurements concerned with the `SMU` are steady state DC measurements. For this reason, prior to making any measurements, a time delay (in seconds) is provided just after the source value is set. During this time interval all transients settle down and steady state equilibrium is achieved. Only when this equilibrium has been achieved do any measurements occur. The default time delay is 2 seconds.

2. Filter Length : The `SMU` hardware comes inbuilt with a data averaging filter. For each data point that is plotted by `Qrius`, internally **n** data points are acquired consecutively and the average of those **n** points is returned as a single data point. The filter length determines the value of **n**, and has a default value of 16.

3. High Resistance Measurement Unit

The Ultra High Resistance module (`Hi-RES`) comes into play in the study of current-voltage characteristics of highly resistive and insulating materials (eg. ceramics, polymers). The `Hi-REs` is used along with the `High Resistivity Variable Temperature Insert` which is specifically designed for ultra low leakage current measurements. Measurements of currents as low as 10 femto-Amperes can be made.

Apart from measurement of currents and voltages vs time, the `Hi-RES` can run in the following modes :

1. R-Time : Resistance of the sample is measured with respect to time. The resistance measurement settings can be configured from `Settings` → `Resistance Measurement Settings` in the `High Resistance Measurement Unit` window.

   Parameters

   (a) Voltage Limit : The maximum voltage (in mV) across the sample, above which acquisition is ended.

   (b) Bipolar : When set to `Yes`, the resistance is measured once, after which the source polarity is switched and resistance is measured again. The two resistance values are averaged to cancel out thermoelectric voltages.

2. I-V : Complete I-V curve of the sample is measured and plotted. Depending on the resistance of the sample, the source is automatically switched between current/voltage. The I-V measurement settings can be configured from `Settings` → `I-V Measurement Settings` in the `Source and Measurement Unit` window.

   Parameters

   (a) Voltage Limit : The maximum voltage (in mV) across the sample, above which acquisition is ended.

   (b) Voltage Step Size : The increment in source voltage (in mV) when voltage source mode is active.

   (c) Bipolar : When set to `Yes`, the I-V characteristics are measured both in the 1st and 2nd quadrants of the I-V space. When set to `No`, the measurement takes place only in the 1st quadrant.

Acquisition Settings can be used to control the time delay after excitation (and before acquisition), and the internal averaging of data points. These settings can be configured inside the `High Resistance Measurement Unit` window by clicking on `Settings` → `Acquisition Settings`. The parameters are explained below :

1. Delay : All measurements concerned with the `Hi-RES` are steady state DC measurements. For this reason, prior to making any measurements, a time delay (in seconds) is provided just after the source value is set. During this time interval all transients settle down and steady state equilibrium is achieved. Only when this equilibrium has been achieved do any measurements occur. The default time delay is 2 seconds.

2. Filter Length : The `Hi-RES` hardware comes inbuilt with a data averaging filter. For each data point that is plotted by `Qrius`, internally **n** data points are acquired consecutively and the average of those **n** points is returned as a single data point. The filter length determines the value of **n**, and has a default value of 16.

D. Measurement Mode Settings

`Measurement Mode Settings` integrate the functionality of multiple modules of the `XPLORE` system into one convenient package. The various modes available are :

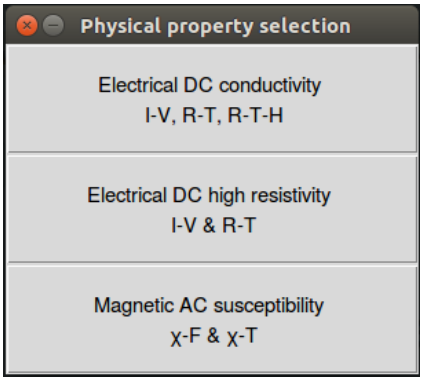


FIG. 32. `Qrius` Measurement Mode Settings

1. Electrical DC Conductivity

This mode integrates the functionality of the `TCON` with the `SMU`, giving the user access to the following run modes for samples with resistances in the range of ($10\ m\Omega\ -\ 100\ M\Omega$) :

1. R-T (Linear Ramp) : The temperature is linearly ramped up, while making intermittent measurements of resistance. The `Linear Ramp Settings` and `Resistance Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).

2. R-T (Stepped Ramp) : The temperature is incremented in steps, and stabilized after each step. The resistance is measured at each step after stabilization. The `Step Ramp Settings` and `Resistance Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).

3. I-V (Stepped Ramp) : The temperature is incremented in steps, and stabilized after each step. The full I-V curve is measured at each step after stabilization. The `Step Ramp Settings` and `I-V Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).

2. Electrical DC High Resistivity

This mode integrates the functionality of the `TCON` with the `SMU`, giving the user access to the following run modes for samples with resistances in the range of ($1\ M\ \Omega\ -\ 100\ T\Omega$) :

1. R-T (Linear Ramp) : The temperature is linearly ramped up, while making intermittent measurements of resistance. The `Linear Ramp Settings` and `Resistance Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).

2. R-T (Stepped Ramp) : The temperature is incremented in steps, and stabilized after each step. The resistance is measured at each step after stabilization. The `Step Ramp Settings` and `Resistance Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).

3. I-V (Stepped Ramp) : The temperature is incremented in steps, and stabilized after each step. The full I-V curve is measured at each step after stabilization. The `Step Ramp Settings` and `I-V Measurement Settings` need to be programmed by the user prior to the run (ref. sections VIII C 1 and VIII C 2).